



The extension problem of the mean curvature flow (I)

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Abstract We show that the mean curvature blows up at the first finite singular time for a closed smooth embedded mean curvature flow in $\mathbb{R}^3$.

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1 Introduction

Let $\mathbf{x}_0 : \Sigma^n \rightarrow \mathbb{R}^{n+1}$ be a closed smooth embedded hypersurface in $\mathbb{R}^{n+1}$. A one-parameter family of immersions $\mathbf{x}(p, t) : \Sigma^n \rightarrow \mathbb{R}^{n+1}$ is called a mean curvature flow, if $\mathbf{x}$ satisfies the equation

$$\frac{\partial \mathbf{x}}{\partial t} = -H \mathbf{n}, \quad \mathbf{x}(0) = \mathbf{x}_0, \quad (1.1)$$

where H denotes the mean curvature of the hypersurface $\Sigma_t := \mathbf{x}(t)(\Sigma)$ and $\mathbf{n}$ denotes the unit normal vector field of Σ_t . In [25], Huisken proved that if the flow (1.1) develops a singularity at time $T < \infty$, then the fundamental form will blow up at time T . A natural conjecture is that the mean curvature will blow up at the finite singular time of a mean curvature flow. There are many results toward this conjecture (c.f. Cooper [4], Le-Sesum [33–35], Lin-Sesum [36], Xu-Ye-Zhao [50] ...). This conjecture is also proposed in page 42 of Mantegazza's book [38] as an open problem. In this paper, we confirm this conjecture in dimension two:

Theorem 1.1 *If $\mathbf{x}(p, t) : \Sigma^2 \rightarrow \mathbb{R}^3 (t \in [0, T))$ is a closed smooth embedded mean curvature flow with the first singular time $T < +\infty$, then*

$$\sup_{\Sigma \times [0, T)} |H|(p, t) = +\infty.$$

It is interesting to compare the extension problem of mean curvature flow with Ricci flow. In [22] Hamilton proved that the Riemann curvature tensor will blow up at the finite singular time of a Ricci flow. In [39] Sesum extended Hamilton's result to the Ricci curvature by using Perelman's noncollapsing theorem. In a series of papers [12, 41, 42] Wang and Chen-Wang provided several conditions which can be used to extend Ricci flow, and they gave a different proof of Sesum's result. It is also conjectured that the Ricci flow can be extended if the scalar curvature stays bounded. Important progresses have been made by Zhang [51], Bamler-Zhang [1] and Simon [40].

The mean curvature flow with convexity conditions has been well studied during the past several decades. If the initial hypersurface satisfies some convexity conditions, like mean convex or two convex, then the mean curvature flow has some convexity estimates (c.f. Huisken [25], Huisken-Sinestrari

[27, 28], Haslhofer-Kleiner [23]). These convexity estimates are important for studying the surgery of mean curvature flow (c.f. Huisken-Sinestrari [29], Brendle-Huisken [2], Haslhofer-Kleiner [24]). In [45, 46] White also gave some important properties of the singularities of a mean curvature flow with mean convex initial hypersurfaces. However, all these results rely on the convexity condition of initial hypersurfaces, and it is very difficult to study general cases (c.f. Colding-Minicozzi [16, 17], T. Ilmanen [30]). Theorem 1.1 can be viewed as an attempt to study the general singularities without assuming convexity conditions.

Now we sketch the proof of Theorem 1.1. Assume that the mean curvature is bounded along the flow (1.1) and the first singular time $T < +\infty$. Consider the corresponding rescaled mean curvature flow

$$\left(\frac{\partial \mathbf{x}}{\partial t}\right)^\perp = -\left(H - \frac{1}{2}\langle \mathbf{x}, \mathbf{n} \rangle\right)\mathbf{n}, \quad \forall t \in [0, \infty). \quad (1.2)$$

Then the mean curvature decays exponentially to zero along the flow (1.2). We have to show that the flow (1.2) converges smoothly to a plane with multiplicity one. The proof consists of two steps:

Step 1. Convergence of the rescaled mean curvature flow with multiplicities.

In this step, we first follow the ideas of Chen-Wang [7, 8] to develop the weak compactness theory of mean curvature flow with certain properties, which are basically area doubling property together with bounded mean curvature H and bounded energy $\int_\Sigma |A|^2$. Note that the energy $\int_\Sigma |A|^2$ can be bounded by H and topology, via the Gauss–Bonnet theorem (c.f. (4.5)). To prove such weak compactness, there are two main technical ingredients:

- The two-sided, long-time pseudolocality theorem.
- The energy concentration property.

The short-time forward pseudolocality theorem for mean curvature flow was studied under the local graphic condition, which says that the surface can be written locally as a graph of a single-valued function. See Ecker-Huisken [19, 20], M.T. Wang [44], Chen-Yin [3] and S. Brendle [2]. In our case, we need to remove this graphic condition since the flow may converge with multiplicities. Under the assumption that the mean curvature is bounded, we will show a new short-time, two-sided pseudolocality type theorem (c.f. Theorem 3.7). Moreover, the pseudolocality theorem can be improved as long-time version (c.f. Theorem 3.8) whenever the mean curvature is very small. Then the energy concentration property follows from the pseudolocality theorem. Once we have the long-time, two-sided pseudolocality theorem and the energy concentration property (c.f. Lemma 3.10), we can show the weak compactness of mean curvature flow and get the “flow” convergence of the rescaled mean curvature flow. Since the limit is both minimal and self-shrinking, we obtain

the limit must be a plane passing through the origin, with possibly more than one multiplicity.

Step 2. Show that the multiplicity of the convergence is one.

In this step, we use the ideas from the compactness of self-shrinkers and minimal surfaces (c.f. Colding-Minicozzi [14, 15], L. Wang [43]) to show that the multiplicity is one. For otherwise we can construct a sequence of positive solutions to the corresponding parabolic equations on compact sets of the limit surface away from singularities. However, we shall show that the existence of such a solution contradicts the fact that the limit is a plane with multiplicities. To obtain the contradiction, there are also two main technical ingredients:

- Uniform estimates of the sequence of positive solutions.
- The L -stability of the limit surface across the singular set.

To show uniform estimates of these positive solutions, we introduce a new “almost” decreasing quantity along the rescaled mean curvature flow, and use this quantity with the parabolic Harnack inequality to control the positive solutions. After normalization and taking the limit, we get a positive solution with good estimates on the limit surface away from singularities. Using these estimates, we show that the limit surface is L -stable away from singularities. Recall that a hypersurface Σ is called L -stable (c.f. Colding-Minicozzi [15]) if for any compactly supported function u , we have

$$\int_{\Sigma} -uLu e^{-\frac{|x|^2}{4}} \geq 0,$$

where L is the operator $Lu = \Delta u + |A|^2 u - \frac{1}{2} \langle x, \nabla u \rangle + \frac{u}{2}$ defined by Colding-Minicozzi [16]. Following the argument of Gulliver-Lawson [21] in minimal surfaces, we show that the limit surface is actually L -stable across the singular set. However, the limit surface is a plane which is not L -stable. Thus we obtain the desired contradiction. This contradiction forces that the convergence of (1.2) must have multiplicity one and hence in the smooth topology.

Once the rescaled mean curvature flow (1.2) converges smoothly to a plane with multiplicity one, we use Huisken’s monotonicity formula and White’s regularity theorem to show that the mean curvature flow (1.1) actually has no singular points at time T . This finishes the proof of Theorem 1.1.

The strategy of the proof of Theorem 1.1 is very similar to the one used in the study of the Kähler Ricci flow on Fano manifolds by Chen-Wang [8] (See [9] and [10] for the published version) and Chen-Sun-Wang [6]. Actually, in Chen-Wang [8], the flow weak-compactness was setup for the normalized Kähler Ricci flow, which is comparable to the step 1 mentioned above. Then in Chen-Sun-Wang [6], the limit and convergence topology can be improved to be “smooth” whenever the underlying manifold is K -stable. This is related to our step 2 described above. However, here we used the L -stability of the

limit surfaces, instead of the K -stability of the the underlying manifolds to rule out the possible singularities. Not surprisingly, the strategy can also be applied to study the extension problem for the 4-dimensional Ricci flow with bounded scalar curvature. Actually, based on the calculation of M. Simon(c.f. [40]), we have uniform bounded energy $\int_M |Rm|^2$ along the flow. Then the weak-compactness of the parabolic normalized Ricci flows (compared with Eq. (1.2))

$$\partial_t g = -2(Ric - g)$$

follows from Chen-Wang [7] (See also [1] for a different approach). The limit of the above flow is then a Ricci-flat gradient shrinking soliton with finite singular points, which is nothing but a flat metric cone over S^3/Γ for some finite subgroup Γ of $O(4)$. Similar to the step 2 mentioned above, the 4-dimensional Ricci flow extension problem will be confirmed if one can develop methods to rule out the nontrivial Γ 's.

We remark that the weak compactness of the mean curvature flow, i.e., step 1, is based on the observation that locally the structure of the mean curvature flow with bounded H and energy $\int_\Sigma |A|^2$ is modeled by the structure of minimal surfaces with uniformly bounded topology. Similar observation is also the key for the weak compactness of the Ricci flow with bounded energy $\int_M |Rm|^{\frac{m}{2}}$ and bounded scalar curvature R . The two-sided, long-time pseudo-locality(c.f. Chen-Wang [7], [8] and [11]) and the energy concentration are the technical tools for writing down the observation in rigorous analysis estimates. It seems that such observation holds for many other geometric flows, e.g., the harmonic map flow and the Calabi flow.

The organization of this paper is as follows. In Sect. 2 we recall some basic facts on mean curvature flow and minimal surfaces. In Sect. 3 we develop the weak compactness theory of mean curvature flow under some geometric conditions. In Sect. 4 we show the rescaled mean curvature flow with the exponential decay of mean curvature converges smoothly to a plane with multiplicity one. Finally, we finish the proof of Theorem 1.1 in Sect. 5.

2 Preliminaries

Let $\mathbf{x}(p, t) : \Sigma^n \rightarrow \mathbb{R}^{n+1}$ be a family of smooth embeddings in $\mathbb{R}^{n+1}$. $\{(\Sigma^n, \mathbf{x}(t)), 0 \leq t < T\}$ is called a mean curvature flow if $\mathbf{x}(t)$ satisfies

$$\frac{\partial \mathbf{x}}{\partial t} = -H\mathbf{n}, \quad \forall t \in [0, T), \quad (2.1)$$

and called a rescaled mean curvature flow if $\mathbf{x}(t)$ satisfies

$$\left(\frac{\partial \mathbf{x}}{\partial t}\right)^\perp = -\left(H - \frac{1}{2}\langle \mathbf{x}, \mathbf{n} \rangle\right) \mathbf{n}, \quad \forall t \in [0, T), \quad (2.2)$$

Sometimes we also write $\mathbf{x}_t$ as $\mathbf{x}(t)$ for short. It is easy to check that the rescaled mean curvature flow is equivalent to the mean curvature flow after rescalings in space and a reparameterization of time. We denote by $B_r(p)$ the open ball in $\mathbb{R}^3$ centered at p with radius r .

It is well known that the volume ratio is bounded from below along the flow (2.1). See, for example, Lemma 2.9 of Colding-Minicozzi [16].

Lemma 2.1 *Let $\{(\Sigma^n, \mathbf{x}(t)), 0 \leq t < T\}$ be a mean curvature flow (2.1). Then there is a constant $N = N(\text{Vol}(\Sigma_0), T) > 0$ such that for all $r > 0$ and $p_0 \in \mathbb{R}^{n+1}$ we have*

$$\text{Vol}(B_r(p_0) \cap \Sigma_t) \leq Nr^n, \quad \forall t \in [0, T).$$

A hypersurface $\mathbf{x} : \Sigma^n \rightarrow \mathbb{R}^{n+1}$ is called a self-shrinker, if it satisfies the equation

$$H = \frac{1}{2}\langle \mathbf{x}, \mathbf{n} \rangle. \quad (2.3)$$

By Corollary 2.8 of Colding-Minicozzi [16], we have

Lemma 2.2 (Corollary 2.8 of [16]) *If Σ is a self-shrinker and the mean curvature is zero, then Σ is a minimal cone. In particular, if Σ is also smooth and embedded, then it is a hyperplane through 0.*

Observe that the Eq. (2.1) is invariant under the rescaling

$$\tilde{\mathbf{x}}(p, s) = \lambda \left(\mathbf{x}(p, T + \frac{s}{\lambda^2}) - p_0 \right), \quad \forall (p, s) \in \Sigma \times [-T\lambda^2, 0), \quad (2.4)$$

where $\lambda > 0$. In fact, under the rescaling (2.4) we have the relations

$$\tilde{A}_{ij} = \lambda A_{ij}, \quad \tilde{g}_{ij} = \lambda^2 g_{ij}, \quad \tilde{H} = \frac{1}{\lambda} H, \quad |\tilde{A}| = \frac{1}{\lambda} |A|,$$

where $\tilde{A}$, $\tilde{g}$ and $\tilde{H}$ denote the second fundamental form, the induced metric and the mean curvature of the surface $\tilde{\Sigma} = \tilde{\mathbf{x}}(\Sigma)$ respectively. Moreover, by direct calculation we have

Lemma 2.3 *Let Σ^2 be a smooth surface in $\mathbb{R}^3$. The area ratio and the L^2 norm of the second fundamental form are invariant under (2.4). Namely, we have*

$$\frac{\text{Area}_g(B_r(p) \cap \Sigma)}{\pi r^2} = \frac{\text{Area}_{\tilde{g}}(B_{\tilde{r}}(\tilde{p}) \cap \tilde{\Sigma})}{\pi \tilde{r}^2}$$

and

$$\int_{B_r(p) \cap \Sigma} |A|^2 d\mu = \int_{B_{\tilde{r}}(\tilde{p}) \cap \tilde{\Sigma}} |\tilde{A}|^2 d\tilde{\mu},$$

where $\tilde{r} = \lambda r$, $\tilde{p} = \lambda(p - p_0)$ and $\tilde{\Sigma} = \lambda(\Sigma - p_0)$.

Now we recall some facts on compactness of immersed hypersurfaces in $\mathbb{R}^{n+1}$. For any $B_R(p) \subset \mathbb{R}^{n+1}$, we say an immersed (or embedded) hypersurface $\Sigma^n \subset \mathbb{R}^{n+1}$ properly immersed (or embedded) in $B_R(p)$, if either Σ is closed or $\partial \Sigma$ has distance at least R from the point p . We say that a sequence $\{\Sigma_i\}$ converges in $C_{\text{loc}}^{k,\alpha}$ topology to a hypersurface Σ , if for any $p \in \Sigma$ each Σ_i is locally (near p) a graph over the tangent space $T_p \Sigma$ and the graph of Σ_i converges to the graph of Σ in the usual $C_{\text{loc}}^{k,\alpha}$ topology.

The following theorem is well known and it shows the convergence of properly immersed surfaces under the second fundamental form bound. The readers are referred to [49] for the details.

Theorem 2.4 (Compactness of minimal surfaces). *Let $\{\Sigma_i^n\}$ be a sequence of smooth, properly immersed hypersurfaces in $B_R(p) \subset \mathbb{R}^{n+1}$. If the second fundamental form $\sup_{\Sigma_i \cap B_R(p)} |A_i| \leq \Lambda$ for a constant $\Lambda > 0$ and all $i \geq 1$, then there is a subsequence of $\{\Sigma_i^n\}$ converges in $C^{1,\frac{1}{2}}$ topology, possibly with multiplicities, to an immersed hypersurface Σ_∞ . Moreover, if the hypersurfaces $\{\Sigma_i^n\}$ are minimal, then the convergence is in smooth topology.*

To study the convergence of mean curvature flow, we define the convergence of a sequence of one-parameter hypersurfaces as follows.

Definition 2.5 We say that a sequence of one-parameter smoothly immersed hypersurfaces $\{\Sigma_{i,t}^n, -1 < t < 1\}$ in $\mathbb{R}^{n+1}$ converges in smooth topology, possibly with multiplicities, to a limit flow $\{\Sigma_{\infty,t}, -1 < t < 1\}$ away from a space-time singular set $\mathcal{S} \subset \mathbb{R}^{n+1} \times (-1, 1)$, if for any $t \in (-1, 1)$, any $p \in \Sigma_{\infty,t} \setminus \mathcal{S}_t$ and large i , there exists $r > 0$ and $\epsilon > 0$ such that the hypersurface $\Sigma_{i,s} \cap B_r(p)$ with $s \in [t - \epsilon, t + \epsilon]$ can be written as a collection of graphs of smooth functions $\{u_i^1(x, s), u_i^2(x, s), \dots, u_i^N(x, s)\}$ over the tangent plane of $\Sigma_{\infty,t}$ at the point p . Moreover, for each $k \in \{1, 2, \dots, N\}$ the functions $u_i^k(x, s)$ converges smoothly in x and s as $i \rightarrow +\infty$.

In the above definition, $\mathcal{S}_t$ is defined by $\mathcal{S}_t = \{x \in \mathbb{R}^{n+1} \mid (x, t) \in \mathcal{S}\}$. If $\mathcal{S}_t$ is independent of t , then we can also replace the space–times singular set $\mathcal{S}$ simply by $\mathcal{S}_{t_0}$ for some t_0 .

The following compactness result of mean curvature flow is well-known. See, for example, page 481–482 of [3] for a detailed proof.

Theorem 2.6 (Compactness of mean curvature flow). *Let $\{(\Sigma_i^n, \mathbf{x}_i(t)), -1 < t < 1\}$ be a sequence of mean curvature flow properly immersed in $B_R(0) \subset \mathbb{R}^{n+1}$ (i.e. for each $t \in (-1, 1)$ the hypersurface $\Sigma_{i,t}$ is properly immersed in $B_R(0)$). Suppose that*

$$\sup_{\Sigma_{i,t} \cap B_R(0)} |A|(x, t) \leq \Lambda, \quad \forall t \in (-1, 1)$$

for some $\Lambda > 0$. Then a subsequence of $\{\Sigma_{i,t} \cap B_R(0), -1 < t < 1\}$ converges in smooth topology to a smooth mean curvature flow $\{\Sigma_{\infty,t}, -1 < t < 1\}$ in $B_R(0)$.

3 Weak compactness of the mean curvature flow

In this section, we follow the arguments of Ricci flow by Chen-Wang in [7, 8] and minimal surfaces by Choi-Schoen in [5] to study the weak compactness of mean curvature flow under some geometric conditions. This weak compactness result will be used to prove the convergence of rescaled mean curvature flow in the next section.

3.1 The pseudolocality theorem

The pseudolocality type results of the mean curvature flow were studied by Ecker-Huisken [19, 20], M. T. Wang [44], Chen-Yin [3] and Brendle-Huisken [2]. However, all these pseudolocality theorems above require the condition that the initial hypersurface can be locally written as a graph of a single-valued function. In our case, we need to remove this graphic condition since the flow may converge with multiplicities. Here, we give a different type of pseudolocality theorem under the assumption that the mean curvature along the flow is uniformly bounded.

Definition 3.1 For any $r > 0$, $p \in \mathbb{R}^{n+1}$ and $\Sigma^n \subset \mathbb{R}^{n+1}$, we denote by $C_x(B_r(p) \cap \Sigma)$ the connected component of $B_r(p) \cap \Sigma$ containing $x \in \Sigma$.

Lemma 3.2 (c.f. Lemma 7.1 of [3]) *Let $\Sigma^n \subset \mathbb{R}^{n+1}$ be properly embedded in $B_{r_0}(x_0)$ for some $x_0 \in \Sigma$ with*

$$|A|(x) \leq \frac{1}{r_0} \quad x \in B_{r_0}(x_0) \cap \Sigma.$$

Let $\{x^1, \dots, x^{n+1}\}$ be the standard coordinates in $\mathbb{R}^{n+1}$. Assume that $x_0 = 0$ and the tangent plane of Σ at x_0 is $x^{n+1} = 0$. Then there is a map

$$u : \left\{ x' = (x^1, \dots, x^n) \mid |x'| < \frac{r_0}{96} \right\} \rightarrow \mathbb{R}$$

with $u(0) = 0$ and $|\nabla u|(0) = 0$ such that the connected component containing x_0 of $\Sigma \cap \{(x', x^{n+1}) \in \mathbb{R}^{n+1} \mid |x'| < \frac{r_0}{96}\}$ can be written as a graph $\{(x', u(x')) \mid |x'| < \frac{r_0}{96}\}$ and

$$|\nabla u|(x') \leq \frac{36}{r_0} |x'|.$$

Using Lemma 3.2, we show that the local area ratio of the surface is very close to 1.

Lemma 3.3 *Suppose that $\Sigma^n \subset B_{r_0}(p) \subset \mathbb{R}^{n+1}$ is a hypersurface with $\partial \Sigma \subset \partial B_{r_0}(p)$ and*

$$\sup_{\Sigma} |A| \leq \frac{1}{r_0}.$$

For any $\delta > 0$, there is a constant $\rho_0 = \rho_0(r_0, \delta)$ such that for any $r \in (0, \rho_0)$ and any $x \in B_{\frac{r_0}{2}}(p) \cap \Sigma$ we have

$$\frac{\text{Vol}_{\Sigma}(C_x(B_r(x) \cap \Sigma))}{\omega_n r^n} \leq 1 + \delta. \quad (3.1)$$

Proof By Lemma 3.2, for any $x \in B_{\frac{r_0}{2}}(p) \cap \Sigma$ the component $C_x(B_{\rho_0}(x) \cap \Sigma)$ with $\rho_0 = \frac{r_0}{192}$ can be written as a graph of a function u over the tangent plane at x , which we assume to be $P = \{(x_1, \dots, x_n, x_{n+1}) \in \mathbb{R}^{n+1} \mid x_{n+1} = 0\}$, with $|\nabla u|(x') \leq \frac{72}{r_0} |x'|$ where $x' = (x_1, \dots, x_n)$. For any $r \in (0, \rho_0)$, the volume ratio of $C_x(B_r(x) \cap \Sigma)$ is given by

$$\frac{\text{Vol}_{\Sigma}(C_x(B_r(x) \cap \Sigma))}{\omega_n r^n} \leq \frac{1}{\omega_n r^n} \int_{B_r(x) \cap P} \sqrt{1 + |\nabla u|^2} d\mu \leq \sqrt{1 + \frac{5184}{r_0^2} r^2}.$$

Thus, we can choose r sufficiently small such that (3.1) holds. The lemma is proved. $\square$

The next result shows that we can control the local volume ratio along the mean curvature flow with bounded mean curvature.

Lemma 3.4 *Let $\{(\Sigma^n, \mathbf{x}(t)), -T \leq t \leq T\}$ be a smooth embedded mean curvature flow (2.1) with $\max_{\Sigma_t \times [-T, T]} |H(p, t)| \leq \Lambda$. Then for any $t_1, t_2 \in [-T, T]$ we have*

$$\frac{\text{Vol}_{g(t_2)}(C_{p_{t_2}}(B_{r_2}(p_{t_2}) \cap \Sigma_{t_2}))}{\omega_n r_2^n} \leq f(t_1, t_2, \Lambda, r_2) \frac{\text{Vol}_{g(t_1)}(C_{p_{t_1}}(B_{r_1}(p_{t_1}) \cap \Sigma_{t_1}))}{\omega_n r_1^n} \quad (3.2)$$

where $p_t = \mathbf{x}_t(p, t)$ for some $p \in \Sigma$ and

$$r_1 = r_2 + 2\Lambda|t_2 - t_1|, \\ f(t_1, t_2, \Lambda, c, r_2) = e^{\Lambda^2|t_2 - t_1|} \left(1 + \frac{2\Lambda}{r_2}|t_2 - t_1|\right)^n$$

Proof Since the mean curvature is bounded, for any $\Omega \subset \Sigma$ we calculate

$$\left| \frac{d}{dt} \text{Vol}_{g(t)}(\Omega) \right| = \left| \int_{\Omega} |H|^2 d\mu_t \right| \leq \Lambda^2 \text{Vol}_{g(t)}(\Omega), \quad (3.3)$$

which implies that

$$e^{-\Lambda^2|t_1 - t_2|} \text{Vol}_{g(t_2)}(\Omega) \leq \text{Vol}_{g(t_1)}(\Omega) \leq e^{\Lambda^2|t_1 - t_2|} \text{Vol}_{g(t_2)}(\Omega). \quad (3.4)$$

Note that for any $p, q \in \Sigma$, we have

$$\begin{aligned} -4\Lambda|\mathbf{x}(p, t) - \mathbf{x}(q, t)| &\leq \frac{\partial}{\partial t} |\mathbf{x}(p, t) - \mathbf{x}(q, t)|^2 \\ &= 2\langle \mathbf{x}(p, t) - \mathbf{x}(q, t), -H(p, t)\mathbf{n}(p, t) + H(q, t)\mathbf{n}(q, t) \rangle \\ &\leq 4\Lambda|\mathbf{x}(p, t) - \mathbf{x}(q, t)|. \end{aligned}$$

Therefore, we have

$$-2\Lambda \leq \frac{\partial}{\partial t} |\mathbf{x}(p, t) - \mathbf{x}(q, t)| \leq 2\Lambda.$$

It follows that

$$|\mathbf{x}(p, t_1) - \mathbf{x}(q, t_1)| \leq |\mathbf{x}(p, t_2) - \mathbf{x}(q, t_2)| + 2\Lambda|t_1 - t_2|. \quad (3.5)$$

Let $p_t = \mathbf{x}(p, t)$. Then we have

$$\mathbf{x}(t_2)^{-1}(C_{p_{t_2}}(B_r(p_{t_2}) \cap \Sigma_{t_2})) \subset \mathbf{x}(t_1)^{-1}(C_{p_{t_1}}(B_{r+2\Lambda|t_1-t_2|}(p_{t_1}) \cap \Sigma_{t_1})).$$

Therefore, we have the estimates

$$\begin{aligned} & \text{Vol}_{g(t_2)}(C_{p_{t_2}}(B_r(p_{t_2}) \cap \Sigma_{t_2})) \\ & \leq \text{Vol}_{g(t_2)}(\mathbf{x}(t_1)^{-1}(C_{p_{t_1}}(B_{r+2\Lambda|t_1-t_2|}(p_{t_1}) \cap \Sigma_{t_1}))) \\ & \leq e^{\Lambda^2|t_1-t_2|} \text{Vol}_{g(t_1)}(C_{p_{t_1}}(B_{r+2\Lambda|t_1-t_2|}(p_{t_1}) \cap \Sigma_{t_1})), \end{aligned} \quad (3.6)$$

where we used (3.4) in the last inequality. For any $x \in \Sigma^n \subset \mathbb{R}^{n+1}$ and $r_2 > 0$, we have

$$\begin{aligned} & \frac{\text{Vol}_{g(t_2)}(C_{p_{t_2}}(B_{r_2}(p_{t_2}) \cap \Sigma_{t_2}))}{\omega_n r_2^n} \\ & \leq e^{\Lambda^2|t_1-t_2|} \cdot \left(\frac{r_1}{r_2}\right)^n \cdot \frac{\text{Vol}_{g(t_1)}(C_{p_{t_1}}(B_{r_1}(p_{t_1}) \cap \Sigma_{t_1}))}{\omega_n r_1^n} \\ & = e^{\Lambda^2|t_1-t_2|} \cdot \left(\frac{r_1}{r_2}\right)^n \cdot \frac{\text{Vol}_{g(t_1)}(C_{p_{t_1}}(B_{r_1}(p_{t_1}) \cap \Sigma_{t_1}))}{\omega_n r_1^n}, \end{aligned}$$

where r_1, r_2 satisfy the following relations:

$$r_1 = r_2 + 2\Lambda|t_1 - t_2|.$$

The lemma is proved. $\square$

Using the idea of the monotonicity formula of minimal surfaces, we show that the volume ratio is almost monotone if the mean curvature is bounded. See, for example, Proposition 1.12 in Colding-Minicozzi [13].

Lemma 3.5 *Let $\Sigma^n \subset \mathbb{R}^{n+1}$ be a properly embedded hypersurface in $B_{r_0}(x_0)$ with $x_0 \in \Sigma$ and $|H| \leq \Lambda$. Then for any $s \in (0, r_0)$ we have*

$$\frac{\text{Vol}_{\Sigma}(B_s(x_0) \cap \Sigma)}{\omega_n s^n} \leq e^{\Lambda r_0} \cdot \frac{\text{Vol}_{\Sigma}(B_{r_0}(x_0) \cap \Sigma)}{\omega_n r_0^n}.$$

In particular, letting $s \rightarrow 0$ we have

$$\text{Vol}_{\Sigma}(B_r(x_0) \cap \Sigma) \geq e^{-\Lambda r} \omega_n r^n, \quad \forall r \in (0, r_0].$$

Proof Note that the function $f(x) = |x - x_0|$ satisfies the identity

$$\Delta_{\Sigma} f^2 = 2n - 2H \langle x - x_0, \mathbf{n} \rangle.$$

By the Stokes' theorem, we have

$$\begin{aligned} 2n \operatorname{Vol}(\{f \leq s\}) &= \int_{\{f \leq s\}} \Delta_{\Sigma} f^2 + 2 \int_{\{f \leq s\}} H \langle x - x_0, \mathbf{n} \rangle \\ &= 2 \int_{\{f=s\}} |(x - x_0)^T| + 2 \int_{\{f \leq s\}} H \langle x - x_0, \mathbf{n} \rangle. \end{aligned} \quad (3.7)$$

The coarea formula implies that

$$\operatorname{Vol}(\{f \leq s\}) = \int_0^s \int_{\{f=r\}} |\nabla_{\Sigma} f|^{-1}. \quad (3.8)$$

Combining the identities (3.7)–(3.8), we have

$$\begin{aligned} \frac{d}{ds} \left(s^{-n} \operatorname{Vol}(\{f \leq s\}) \right) &= -ns^{-n-1} \operatorname{Vol}(\{f \leq s\}) + s^{-n} \int_{\{f=s\}} \frac{|x - x_0|}{|(x - x_0)^T|} \\ &= s^{-n-1} \int_{\{f=s\}} \frac{|(x - x_0)^N|^2}{|(x - x_0)^T|} - s^{-n-1} \int_{\{f \leq s\}} H \langle x - x_0, \mathbf{n} \rangle \\ &\geq -\Lambda \cdot s^{-n} \operatorname{Vol}(\{f \leq s\}). \end{aligned}$$

Let $F(s) = s^{-n} \operatorname{Vol}(\{f \leq s\})$. Then for any $s \in (0, r_0)$ we have

$$F(s) \leq F(r_0) e^{\Lambda(r_0-s)} \leq F(r_0) e^{\Lambda r_0}.$$

Thus, the lemma is proved. $\square$

Lemma 3.6 *For any $C > 0$, there exists $\delta = \delta(n, C) > 0$ satisfying the following property. Any complete smooth minimal hypersurface $\Sigma^n \subset \mathbb{R}^{n+1}$ with bounded second fundamental form $|A| \leq C$ and volume ratio*

$$\frac{\operatorname{Vol}_{\Sigma}(B_r(p) \cap \Sigma)}{\omega_n r^n} < 1 + \delta, \quad \forall r > 0 \quad (3.9)$$

must be a hyperplane.

Proof Suppose not, there exists a sequence of non-flat minimal hypersurfaces Σ_i with $|A_i| \leq C$ and

$$\frac{\operatorname{Vol}(B_r(p_i) \cap \Sigma_i)}{\omega_n r^n} < 1 + \delta_i, \quad \forall r > 0 \quad (3.10)$$

where $p_i \in \Sigma_i$ and $\delta_i \rightarrow 0$. Since Σ_i are non-flat, we can assume that $|A_i|(p_i) = 1$. By Theorem 2.4, a subsequence of $\tilde{\Sigma}_i = \Sigma_i - p_i$ converges

smoothly to a complete smooth minimal hypersurfaces Σ_∞ with $|A_\infty|(0) = 1$ and volume ratio

$$\frac{\text{Vol}(B_r(0) \cap \Sigma_\infty)}{\omega_n r^n} = 1, \quad \forall r > 0. \quad (3.11)$$

(3.11) implies that Σ_∞ is a hyperplane (c.f. Corollary 1.13 of Colding-Minicozzi [13]), which contradicts the equality $|A_\infty|(0) = 1$. Thus, the lemma is proved. $\square$

Combining the above results, we show the following pseudolocality theorem.

Theorem 3.7 (Two-sided pseudolocality). *For any $r_0 \in (0, 1]$, $\Lambda, T > 0$, there exist $\eta = \eta(n, \Lambda), \epsilon = \epsilon(n, \Lambda) > 0$ satisfying*

$$\lim_{\Lambda \rightarrow 0} \eta(n, \Lambda) = \eta_0(n) > 0, \quad \lim_{\Lambda \rightarrow 0} \epsilon(n, \Lambda) = \epsilon_0(n) > 0 \quad (3.12)$$

and the following properties. Let $\{(\Sigma^n, \mathbf{x}(t)), -T \leq t \leq T\}$ be a closed smooth embedded mean curvature flow (2.1). Assume that

- (1) the second fundamental form satisfies $|A|(x, 0) \leq \frac{1}{r_0}$ for any $x \in C_{p_0}(B_{r_0}(p_0) \cap \Sigma_0)$ where $p_0 = \mathbf{x}_0(p)$ for some $p \in \Sigma$;
- (2) the mean curvature of $\{(\Sigma^n, \mathbf{x}(t)), -T \leq t \leq T\}$ is bounded by Λ .

Then for any (x, t) satisfying

$$x \in C_{p_t}(\Sigma_t \cap B_{\frac{1}{16}r_0}(p_0)), \quad t \in \left[-\frac{\eta r_0^2}{2(\Lambda + \Lambda^2)}, \frac{\eta r_0^2}{2(\Lambda + \Lambda^2)} \right] \cap [-T, T] \quad (3.13)$$

where $p_t = \mathbf{x}_t(p)$, we have the estimate

$$|A|(x, t) \leq \frac{1}{\epsilon r_0}.$$

Proof The proof consists of the following steps:

Step 1. Without loss of generality, we assume $r_0 = 1$. By Lemma 3.3 and the assumption (1), for any fixed $\delta > 0$ there exists a constant $\tilde{\rho}_0 = \tilde{\rho}_0(\delta) \in (0, \frac{1}{2}]$ such that for any $r \in (0, \tilde{\rho}_0]$ we have

$$\frac{\text{Vol}_g(0)(C_{y_0}(B_r(y_0) \cap \Sigma_0))}{\omega_n r^n} \leq 1 + \delta, \quad \forall y_0 \in C_{p_0}(B_{\frac{1}{2}}(p_0) \cap \Sigma_0). \quad (3.14)$$

For any $\rho_0 \in (0, \tilde{\rho}_0]$, we define

$$\eta_0(n, \rho_0, \delta) := \sup \left\{ \eta \in (0, \frac{1}{8}] \mid e^\eta \left(1 + \frac{2\eta}{\rho_0 - 2\eta} \right)^n (1 + \delta) \leq 1 + 2\delta \right\}. \quad (3.15)$$

For any given $\Lambda > 0$, we choose $\rho_0 = \rho_0(n, \delta, \Lambda) \in (0, \tilde{\rho}_0]$ such that

$$\rho_0(n, \delta, \Lambda) := \sup \left\{ \rho_0 \in (0, \tilde{\rho}_0] \mid e^{\Lambda(\rho_0 - 2\eta_0(n, \rho_0, \delta))}(1 + 2\delta) \leq 1 + 3\delta \right\}, \quad (3.16)$$

where η_0 is defined by (3.15). Note that ρ_0 and η_0 have positive lower bounds depending only on n and δ as $\Lambda \rightarrow 0$.

Step 2. By Lemma 3.4, (3.14) and (3.15), for any $t \in [-T, T]$ with $\Lambda|t| + \Lambda^2|t| \leq \eta_0$ we have

$$\begin{aligned} \frac{\text{Vol}_{g(t)}(C_{y_t}(B_{\rho_1}(y_t) \cap \Sigma_t))}{\omega_n \rho_1^n} &\leq e^{\Lambda^2|t|} \left(1 + \frac{2\Lambda|t|}{\rho_1}\right)^n \frac{\text{Vol}_{g(0)}(C_{y_0}(B_{\rho_0}(y_0) \cap \Sigma_0))}{\omega_n \rho_0^n} \\ &\leq 1 + 2\delta, \end{aligned} \quad (3.17)$$

where $y_t = \mathbf{x}_t(\mathbf{x}_0^{-1}(y_0))$, y_0 is any point in $C_{p_0}(B_{\frac{1}{2}}(p_0) \cap \Sigma_0)$ and $\rho_1 := \rho_0 - 2\eta_0(n, \rho_0, \delta)$. Note that $\rho_1 > 0$ by (3.15). Let $\eta_1 = \sqrt{\frac{\eta_0}{2(\Lambda + \Lambda^2)}}$. Then (3.17) holds for any $t \in [-2\eta_1^2, 2\eta_1^2] \cap [-T, T]$. By Lemma 3.5, (3.16), (3.17) and the definition of ρ_1 , for any $s \in (0, \rho_1]$ and any $t \in [-2\eta_1^2, 2\eta_1^2] \cap [-T, T]$ we have

$$\begin{aligned} \frac{\text{Vol}_{g(t)}(C_{y_t}(B_s(y_t) \cap \Sigma_t))}{\omega_n s^n} &\leq e^{\Lambda\rho_1} \frac{\text{Vol}_{g(t)}(C_{y_t}(B_{\rho_1}(y_t) \cap \Sigma_t))}{\omega_n \rho_1^n} \\ &\leq 1 + 3\delta. \end{aligned} \quad (3.18)$$

Let $\bar{\rho} := \frac{1}{4} - \eta_0$. Then by the definition of η_0 we have $\frac{1}{8} \leq \bar{\rho} \leq \frac{1}{4}$. By the assumption (2), we have $p_t \in B_{\bar{\rho}}(p_0) \cap \Sigma_t \neq \emptyset$ for any $t \in [-2\eta_1^2, 2\eta_1^2] \cap [-T, T]$ since

$$|p_t - p_0| \leq \Lambda|t| \leq 2\Lambda\eta_1^2 = \frac{\eta_0}{1 + \Lambda} < \eta_0 \leq \frac{1}{8} \leq \bar{\rho}.$$

Using the assumption (2) again, for any $q_t \in C_{p_t}(B_{\bar{\rho}}(p_0) \cap \Sigma_t)$ with $t \in [-2\eta_1^2, 2\eta_1^2] \cap [-T, T]$ we have $q_0 := \mathbf{x}_0(\mathbf{x}_t^{-1}(q_t)) \in C_{p_0}(B_{\frac{1}{2}}(p_0) \cap \Sigma_0)$ since

$$|q_0 - p_0| \leq |q_0 - q_t| + |q_t - p_0| < \frac{1}{8} + \bar{\rho} \leq \frac{1}{2}.$$

Combining this with (3.18), for any $t \in [-2\eta_1^2, 2\eta_1^2] \cap [-T, T]$ and $q \in C_{p_t}(B_{\bar{\rho}}(p_0) \cap \Sigma_t)$ we have

$$\frac{\text{Vol}_{g(t)}(C_q(B_s(q) \cap \Sigma_t))}{\omega_n s^n} \leq 1 + 3\delta, \quad \forall s \in (0, \rho_1]. \quad (3.19)$$

Step 3. Suppose there exist $\Lambda > 0$, a sequence of $\epsilon \rightarrow 0$, $\epsilon \in (0, \eta_1]$ and smooth solutions to the mean curvature flow $\mathbf{x}_t : \Sigma^n \rightarrow \mathbb{R}^{n+1}$ for $t \in [-T, T]$ with $T \geq 2\eta_1^2$ such that $|A|(x, 0) \leq 1$ for any $x \in C_{p_0}(B_1(p_0) \cap \Sigma_0)$, and there exists (x_1, t_1) satisfying $t_1 \in [-\eta_1^2, \eta_1^2]$ and $x_1 \in C_{p_{t_1}}(B_{\frac{1}{16}}(p_0) \cap \Sigma_{t_1})$ such that

$$Q_1 := |A|(x_1, t_1) > \frac{1}{\epsilon}. \quad (3.20)$$

Note that $p_{t_1} \in B_{\frac{1}{16}}(p_0) \cap \Sigma_{t_1}$ since

$$|p_{t_1} - p_0| \leq \Lambda \eta_1^2 \leq \frac{\eta_0}{2(1 + \Lambda)} \leq \frac{1}{16}.$$

Fix $K > 0$ such that $K > \frac{1}{2}\Lambda\epsilon$ and $2K\epsilon < \frac{1}{16}$. Check whether there exists a point

$$(x, t) \in C_{x_{1,t}}(B_{KQ_1^{-1}}(x_1) \cap \Sigma_t) \times [t_1 - \frac{1}{2}Q_1^{-2}, t_1] \quad (3.21)$$

satisfying $|A|(x, t) > 2Q_1$. Here we define $x_{1,t} = \mathbf{x}_t(\mathbf{x}_{t_1}^{-1}(x_1))$. Note that $x_{1,t} \in B_{KQ_1^{-1}}(x_1) \cap \Sigma_t$ since

$$|x_1 - x_{1,t}| \leq \Lambda|t| \leq \frac{1}{2}\Lambda Q_1^{-2} < KQ_1^{-1}, \quad \forall t \in [t_1 - \frac{1}{2}Q_1^{-2}, t_1].$$

If there is no such point, then we stop. Otherwise, we can find a point, which we denote by (x_2, t_2) , satisfying (3.21) and $Q_2 := |A|(x_2, t_2) > 2Q_1$. Then we check whether there exists a point

$$(x, t) \in C_{x_{2,t}}(B_{KQ_2^{-1}}(x_2) \cap \Sigma_t) \times [t_2 - \frac{1}{2}Q_2^{-2}, t_2] \quad (3.22)$$

satisfying $|A|(x, t) > 2Q_2$. We can also check that $x_{2,t} \in B_{KQ_2^{-1}}(x_2) \cap \Sigma_t$. If there is no such point, then we stop. Otherwise, we can find a point which we denote by (x_3, t_3) . Repeating the process, we can find a sequence of points (x_k, t_k) . Note that

$$t_k \geq t_1 - \frac{1}{2}(Q_{k-1}^{-2} + Q_{k-2}^{-2} + \cdots + Q_1^{-2}) \geq t_1 - \epsilon^2 > -2\eta_1^2$$

and the Euclidean distance

$$\begin{aligned}
 d(x_k, p_0) &\leq d(x_k, x_{k-1}) + d(x_{k-1}, x_{k-2}) + \cdots + d(x_1, p_0) \\
 &\leq K \left(Q_{k-1}^{-1} + Q_{k-2}^{-1} + \cdots + Q_1^{-1} \right) + d(x_1, p_0) \\
 &\leq 2K\epsilon + \frac{1}{16} \\
 &\leq \frac{1}{8} \leq \bar{\rho},
 \end{aligned}$$

where we choose $K = \epsilon^{-\frac{1}{2}}$ and ϵ small. Since $Q_k := |A|(x_k, t_k) \geq 2^{k-1} Q_1 \rightarrow +\infty$ as $k \rightarrow +\infty$, the process will stop at some finite k and we get a point $(\bar{x}, \bar{t})$ satisfying the following properties:

- $(\bar{x}, \bar{t}) \in C_{p_{\bar{t}}}(B_{\bar{\rho}}(p_0) \cap \Sigma_{\bar{t}}) \times (-2\eta_1^2, t_1]$;
- $|A|(x, t) \leq 2|A|(\bar{x}, \bar{t})$ for any point $(x, t) \in C_{\bar{x}, t}(B_{K\bar{Q}^{-1}}(\bar{x}) \cap \Sigma_t) \times [\bar{t} - \frac{1}{2}\bar{Q}^{-2}, \bar{t}]$, where $\bar{x}_t := \mathbf{x}_t(\mathbf{x}_{\bar{t}}^{-1}(\bar{x}))$ and $\bar{Q} := |A|(\bar{x}, \bar{t})$. Note that $[\bar{t} - \frac{1}{2}\bar{Q}^{-2}, \bar{t}] \subset [-2\eta_1^2, t_1]$.

Step 4. We rescale the flow by

$$\tilde{\mathbf{x}}(p, s) = \bar{Q} \left(\mathbf{x}(p, \bar{t} + \frac{s}{\bar{Q}^2}) - \bar{x} \right), \quad \forall (p, s) \in \Sigma \times [-(T + \bar{t})\bar{Q}^2, (T - \bar{t})\bar{Q}^2].$$

Then the rescaled flow $\tilde{\Sigma}_s := \tilde{\mathbf{x}}_s(\Sigma)$ is a mean curvature flow satisfying the following properties:

- For any $(x, s) \in C_{\tilde{\mathbf{x}}_s(\tilde{\mathbf{x}}_0^{-1}(0))}(B_{\epsilon^{-\frac{1}{2}}}(0) \cap \tilde{\Sigma}_s) \times [-\frac{1}{2}, 0]$, we have $|A_{\tilde{\Sigma}_s}|(x, s) \leq 2$ and $|A_{\tilde{\Sigma}_0}|(0, 0) = 1$;
- For any $r \in (0, \epsilon^{-\frac{1}{2}})$ we have the volume ratio

$$\frac{\text{Vol}(C_0(B_r(0) \cap \tilde{\Sigma}_0))}{\omega_n r^n} \leq 1 + 3\delta.$$

Here we used (3.19), (3.20) and the facts that $K\bar{Q}^{-1} \leq KQ_1^{-1} \leq \epsilon^{\frac{1}{2}} < \rho_1$ when ϵ is small.

- The mean curvature of the flow $\tilde{\Sigma}_s$ satisfies $|\tilde{H}| \leq \Lambda \bar{Q}^{-1}$.

Since $\epsilon \rightarrow 0$ and $\bar{Q} \rightarrow +\infty$, the flow $C_{\tilde{\mathbf{x}}_s(\tilde{\mathbf{x}}_0^{-1}(0))}(B_{\epsilon^{-\frac{1}{2}}}(0) \cap \tilde{\Sigma}_s) \times [-\frac{1}{2}, 0]$ converges smoothly to a complete smooth minimal surface Σ_∞ with $\sup_{\Sigma_\infty} |A_{\Sigma_\infty}| \leq 2$, $|A_{\Sigma_\infty}|(0) = 1$ and volume ratio

$$\frac{\text{Vol}(B_r(0) \cap \Sigma_\infty)}{\omega_n r^n} \leq 1 + 3\delta, \quad \forall r > 0.$$

If we choose $\delta = \frac{1}{3}\delta_0$ where $\delta_0 = \delta_0(n)$ is the constant in Lemma 3.6, then Σ_∞ is a hyperplane, which contradicts $|A_{\Sigma_\infty}|(0) = 1$. The theorem is proved. $\square$

A direct corollary of Theorem 3.7 is the following long time pseudolocality theorem. The long-time-pseudolocality type theorem originates from the study of the Kähler Ricci flow by Chen-Wang (c.f. Theorem 1.4 of Chen-Wang [8], or Proposition 4.15 and Remark 5.3 of Chen-Wang [11]). It will be inspiring to compare the following theorem with its Kähler Ricci flow counterpart.

Theorem 3.8 (Long-time, two-sided pseudolocality). *For any $r_0 \in (0, 1]$, $T > 0$, there exist $\delta = \delta(n, r_0, T)$, $\epsilon = \epsilon(n) > 0$ with the following properties. Let $\{(\Sigma^n, \mathbf{x}(t)), -T \leq t \leq T\}$ be a closed smooth embedded mean curvature flow (2.1). Assume that*

- (1) *the second fundamental form satisfies $|A|(x, 0) \leq \frac{1}{r_0}$ for any $x \in C_{p_0}(B_{r_0}(p_0) \cap \Sigma_0)$ where $p_0 = \mathbf{x}_0(p)$ for some $p \in \Sigma$;*
- (2) *the mean curvature of $\{(\Sigma^n, \mathbf{x}(t)), -T \leq t \leq T\}$ is bounded by δ .*

Then for any $(x, t) \in C_{p_t}(\Sigma_t \cap B_{\frac{1}{16}r_0}(p_0)) \times [-T, T]$ where $p_t = \mathbf{x}_t(p)$, we have the estimate

$$|A|(x, t) \leq \frac{1}{\epsilon r_0}.$$

Proof We apply Theorem 3.7 for $\Lambda = \delta$, then we get the constant $\eta(n, \delta)$ and $\epsilon(n, \delta)$. By (3.13), the conclusion holds for any $t \in [-T, T]$ if

$$\frac{\eta(n, \delta)r_0^2}{2(\delta + \delta^2)} \geq 2T. \quad (3.23)$$

Since $\eta_0(n) = \lim_{\delta \rightarrow 0} \eta(n, \delta) > 0$, there exists a constant $\delta = \delta(n, r_0, T)$ such that (3.23) holds. Note that $\lim_{\delta \rightarrow 0} \epsilon(n, \delta) = \epsilon_0(n) > 0$ by (3.12). Thus, the theorem is proved. $\square$

3.2 Energy concentration property

In [5], Choi-Schoen showed the following energy concentration property for minimal surfaces. This property says that the energy near a point with large curvature cannot be small.

Lemma 3.9 (Choi-Schoen [5]) *Fix $\rho \leq 1$. There is a number $\epsilon_0 > 0$ such that if $\Sigma^2 \subset \mathbb{R}^3$ is a minimal surface with $\partial \Sigma \subset \partial B_\rho(x)$ with $|A|(x) \geq \rho^{-1}$, then*

$$\int_{B_{\frac{\rho}{2}}(x) \cap \Sigma} |A|^2 d\mu \geq \epsilon_0.$$

Motivated by Choi-Schoen's result, we show that the energy concentration property holds for mean curvature flow with bounded mean curvature by using the pseudolocality theorem.

Lemma 3.10 (Energy concentration). *For any $\Lambda, K, T > 0$, there exists a constant $\epsilon(n, \Lambda, K, T) > 0$ with the following property. Let $\{(\Sigma^n, \mathbf{x}(t)), -T \leq t \leq T\}$ be a closed smooth embedded mean curvature flow (2.1). Assume that $\max_{\Sigma_t \times [-T, T]} |H|(p, t) \leq \Lambda$. Then we have*

$$\int_{\Sigma_0 \cap B_{Q^{-1}(q)} |A|^n d\mu_0 \geq \epsilon(n, \Lambda, K, T) \quad (3.24)$$

whenever $q \in \Sigma_0$ with $Q := |A|(q, 0) \geq K$.

Proof Let $x_0 \in \Sigma_0$ such that $Q := |A|(x_0, 0) \geq K$. We define the function $f(x) = |A|(x, 0)d(x, \partial\Omega)$ on $\Omega := B_{Q^{-1}(x_0)} \cap \Sigma_0$. Here d denotes the Euclidean distance in $\mathbb{R}^{n+1}$. Note that $f = 0$ on the boundary $\partial\Omega$, if $\partial\Omega \neq \emptyset$, and $f = 1$ at the center point x_0 .

Case 1. $\max_{\Omega} f < 10$. We rescale the flow by $\tilde{\mathbf{x}}(p, t) = Q(\mathbf{x}(p, Q^{-2}t) - x_0)$ and let $\tilde{\Omega} := B_1(0) \cap \tilde{\Sigma}_0$, where $\tilde{\Sigma}_0 := \tilde{\mathbf{x}}(0)(\Sigma)$. Note that the mean curvature of $\tilde{\Sigma}_t$ ($-K^2T \leq t \leq K^2T$) satisfies

$$\max_{\tilde{\Sigma}_t \times [-K^2T, K^2T]} |\tilde{H}|(p, t) \leq \frac{\Lambda}{Q} \leq \frac{\Lambda}{K}, \quad (3.25)$$

and $\max_{\tilde{\Omega}} |\tilde{A}|d(x, \partial\tilde{\Omega}) = \max_{\Omega} f < 10$. Thus, we have $|\tilde{A}|(x, 0) < \frac{10}{d(x, \partial\tilde{\Omega})}$ for any $x \in B_1(0) \cap \tilde{\Sigma}_0$. In particular, inside $B_{\frac{1}{2}}(0) \cap \tilde{\Sigma}_0$ we have $|\tilde{A}| < 20$. Theorem 3.7 implies that there exists $\delta_0 = \delta_0(n, \Lambda, K, T) \in (0, 1)$ such that $|\tilde{A}|(p, t) \leq \frac{1}{\delta_0}$ for any $t \in [-\delta_0^2, \delta_0^2]$ and $p \in \tilde{\Sigma}_t \cap B_{\delta_0}(0)$. Therefore, there exists $\delta_1 = \delta_1(n, \Lambda, K, T) \in (0, \delta_0)$ such that we have all higher order curvature estimates in $B_{\delta_1}(0) \cap \tilde{\Sigma}_t$ for any $t \in [-\delta_1^2, \delta_1^2]$. Note that $|\tilde{A}|(0, 0) = 1$, the higher order curvature estimates implies that $|\tilde{A}|(q, 0) \geq \frac{1}{2}$ on $B_{\delta_2}(0) \cap \tilde{\Sigma}_0$ for some $\delta_2(n, \Lambda, K, T) \in (0, \delta_1)$. Therefore, we have

$$\begin{aligned} \int_{B_{Q^{-1}(x_0)} \cap \Sigma_0} |A|^n d\mu_0 &= \int_{B_1(0) \cap \tilde{\Sigma}_0} |\tilde{A}|^n d\tilde{\mu}_0 \geq \int_{B_{\delta_2}(0) \cap \tilde{\Sigma}_0} |\tilde{A}|^n d\tilde{\mu}_0 \\ &\geq \frac{1}{2^n} \text{Vol}_{\tilde{\Sigma}_0}(B_{\delta_2}(0) \cap \tilde{\Sigma}_0) \\ &\geq \frac{1}{2^n} \cdot \omega_n e^{-\frac{\Lambda\delta_2}{K}} \delta_2^n, \end{aligned}$$

where we used Lemma 3.5 in the last inequality.

Case 2. $\max_{\Omega} f \geq 10$. Let y_0 be the point where f achieve the maximum and $Q' := |A|(y_0, 0)$. Note that

$$Q' \geq \frac{10}{d(y_0, \partial\Omega)} \geq 10Q \geq 10K.$$

We rescale the flow by $\tilde{\mathbf{x}}(p, t) = Q'(\mathbf{x}(p, Q'^{-2}t) - x_0)$ and we define

$$\tilde{y}_0 := Q'(y_0 - x_0), \quad \tilde{\Omega} := B_{Q'Q^{-1}}(0) \cap \tilde{\Sigma}_0,$$

where $\tilde{\Sigma}_0 := \tilde{\mathbf{x}}(0)(\Sigma)$. Then the function $\tilde{f}(x, 0) := |\tilde{A}|(x, 0)d(x, \partial\tilde{\Omega})$ achieves the maximum at the point $\tilde{y}_0$ and the mean curvature of $\tilde{\Sigma}_t$ is bounded by $\frac{\Lambda}{10K}$ for any $t \in [-Q'^2T, Q'^2T]$. Moreover, $|\tilde{A}|(\tilde{y}_0, 0) = 1$ and $d(\tilde{y}_0, \partial\tilde{\Omega}) \geq 10$. For any $x \in B_1(\tilde{y}_0) \cap \tilde{\Sigma}_0$, we have

$$|\tilde{A}|(x, 0) \leq \frac{d(\tilde{y}_0, \partial\tilde{\Omega})}{d(x, \partial\tilde{\Omega})} |\tilde{A}|(\tilde{y}_0, 0) \leq \frac{d(\tilde{y}_0, \partial\tilde{\Omega})}{d(\tilde{y}_0, \partial\tilde{\Omega}) - 1} \leq 2.$$

Then using the backward pseudolocality as in case 1, we obtain that

$$\int_{B_1(\tilde{y}_0) \cap \tilde{\Sigma}_0} |\tilde{A}|^n d\tilde{\mu}_0 \geq \epsilon(n, \Lambda, K, T)$$

for some $\epsilon(n, \Lambda, K, T) > 0$. Using the scaling invariance, we have $\int_{B_{Q'^{-1}}(y_0) \cap \Sigma_0} |A|^n d\mu_0 > \epsilon(n, \Lambda, K, T)$. By the definition of y_0 , we have $Q'^{-1} \leq \frac{1}{10}d(y_0, \partial\Omega)$. Therefore, $B_{Q'^{-1}}(y_0) \subset B_{Q^{-1}}(x_0)$ and we have the inequality

$$\int_{B_{Q^{-1}}(x_0) \cap \Sigma_0} |A|^n d\mu_0 > \epsilon(n, \Lambda, K, T).$$

The lemma is proved. $\square$

A direct corollary of Lemma 3.10 is the following result.

Corollary 3.11 (ϵ -regularity) *There exists $\epsilon_0(n) > 0$ satisfying the following property. Suppose $\{(\Sigma^n, \mathbf{x}(t)), -1 \leq t \leq 1\}$ is a closed smooth embedded mean curvature flow (2.1). Suppose that the mean curvature satisfies $\max_{\Sigma_t \times [-1, 1]} |H|(p, t) \leq 1$. For any $q \in \Sigma_0$, if*

$$\int_{\Sigma_0 \cap B_r(q)} |A|^n d\mu_0 \leq \epsilon_0(n) \tag{3.26}$$

for some $r > 0$, then we have

$$\max_{B_{\frac{r}{2}}(q) \cap \Sigma_0} |A| \leq \max\{1, \frac{2}{r}\}. \quad (3.27)$$

Proof For any $p \in B_{\frac{r}{2}}(q) \cap \Sigma_0$, if $Q := |A|(p, 0)$ satisfies $Q^{-1} < \frac{r}{2}$ and $Q > 1$, then by Lemma 3.10 we have

$$\int_{B_r(q) \cap \Sigma_0} |A|^n d\mu_0 \geq \int_{B_{Q^{-1}}(p) \cap \Sigma_0} |A|^n d\mu_0 \geq \epsilon_0(n),$$

where ϵ_0 is the constant determined by choosing $K = \Lambda = 1$ in Lemma 3.10. Therefore, we have $Q \leq 1$ or $Q^{-1} \geq \frac{r}{2}$, which implies (3.27). The corollary is proved. $\square$

3.3 Weak compactness

In this subsection, we focus on the case $n = 2$. As in Ricci flow [7], we study the refined sequence of mean curvature flow, which can be viewed as a sequence blown up from a rescaled mean curvature flow with bounded mean curvature and bounded energy.

Definition 3.12 (Refined sequences). Let $\{(\Sigma_i^2, \mathbf{x}_i(t)), -1 \leq t \leq 1\}$ be a one-parameter family of closed smooth embedded surfaces satisfying the mean curvature flow equation (2.1). It is called a refined sequence if the following properties are satisfied for every i :

- (1) There exists a constant $D > 0$ such that $d(\Sigma_{i,t}, 0) \leq D$, where $d(\Sigma, 0)$ denotes the Euclidean distance from the point $0 \in \mathbb{R}^3$ to the surface $\Sigma \subset \mathbb{R}^3$.
- (2) The mean curvature satisfies the inequality

$$\lim_{i \rightarrow +\infty} \max_{\Sigma_{i,t} \times [-1, 1]} |H_i|(p, t) = 0. \quad (3.28)$$

- (3) There is a uniform constant Λ such that

$$\int_{\Sigma_{i,t}} |A_i|^2 d\mu_{i,t} \leq \Lambda, \quad \forall t \in [-1, 1]. \quad (3.29)$$

- (4) There is uniform $N > 0$ such that for all $r > 0$ and $p \in \mathbb{R}^3$ we have

$$r^{-2} \text{Area}_{g_i(t)}(B_r(p) \cap \Sigma_{i,t}) \leq N, \quad \forall t \in [-1, 1]. \quad (3.30)$$

- (5) There exist uniform constants $\bar{r}, \kappa > 0$ such that for any $r \in (0, \bar{r}]$ and any $p \in \Sigma_{i,t}$ we have

$$r^{-2} \text{Area}_{g_i(t)}(B_r(p) \cap \Sigma_{i,t}) \geq \kappa, \quad \forall t \in [-1, 1]. \quad (3.31)$$

Proposition 3.13 (Weak compactness of refined sequences). *If $\{(\Sigma_i^2, \mathbf{x}_i(t)), -1 \leq t \leq 1\}$ is a refined sequence in the sense of Definition 3.12, then there exists a finite set of points $S_0 \subset \mathbb{R}^3$ and a smooth embedded minimal surface Σ_∞ such that a subsequence of $\{(\Sigma_i^2, \mathbf{x}_i(t)), -1 < t < 1\}$ converges in smooth topology, possibly with multiplicity at most N_0 , to $\{\Sigma_\infty\}$ away from a finite set S_0 of at most M_0 points. The number N_0 and M_0 can be chosen as*

$$N_0 = \left\lceil \frac{N}{\kappa} \right\rceil + 1, \quad M_0 = \left\lceil \frac{\Lambda}{\epsilon_0} \right\rceil + 1, \quad (3.32)$$

where $\lceil \cdot \rceil$ means the integer part of a nonnegative number, N, κ, Λ and ϵ_0 are the numbers in (3.30), (3.31), (3.29) and (3.26) respectively. Furthermore, the subsequence also converges to Σ_∞ in (extrinsic) Hausdorff distance.

Proof We follow the argument of compactness of minimal surfaces (c.f. White [48, 49], or Colding-Minicozzi [13]). Fix large $\rho > 0$ and let $\Omega = B_\rho(0) \subset \mathbb{R}^3$. By Property (1) in Definition 3.12, we have $\Sigma_{i,0} \cap \Omega \neq \emptyset$ for large ρ . For any $U \subset \Omega$, we define the measures ν_i by

$$\nu_i(U) = \int_{U \cap \Sigma_{i,0}} |A_i|^2 d\mu_{i,0} \leq \Lambda.$$

The general compactness of Radon measures implies that there is a subsequence, which we still denote by ν_i , converges weakly to a Radon measure ν with $\nu(\Omega) \leq \Lambda$. We define the set

$$S_0 = \{x \in \Omega \mid \nu(x) \geq \epsilon_0\},$$

where ϵ_0 is the constant in Corollary 3.11. It follows that S_0 contains at most $\frac{\Lambda}{\epsilon_0}$ points, which is independent of ρ . Given any $y \in \Omega \setminus S_0$. There exists some $s \in (0, \frac{1}{5})$ such that $B_{10s}(y) \subset \Omega$ and $\nu(B_{10s}(y)) < \epsilon_0$. Since $\nu_i \rightarrow \nu$, for i sufficiently large we have

$$\int_{B_{10s}(y) \cap \Sigma_{i,0}} |A_i|^2 d\mu_{i,0} < \epsilon_0.$$

Corollary 3.11 implies that for i sufficiently large we have the estimate

$$\max_{B_{5s}(y) \cap \Sigma_{i,0}} |A|(x, 0) \leq \max\{1, \frac{1}{5s}\} \leq \frac{1}{5s}. \quad (3.33)$$

Note that by Property (2) in Definition 3.12 the mean curvature of $\Sigma_{i,t}$ tends to zero. By Theorem 3.8 there exists a universal constant $\epsilon > 0$ such that for large i and any small $s \in (0, \frac{1}{5})$ we have

$$\max_{B_{\frac{1}{16}r_0}(y) \cap \Sigma_{i,t}} |A|(x, t) \leq \frac{1}{\epsilon r_0}, \quad \forall t \in [-1, 1], \quad (3.34)$$

where $r_0 = 5s$. Therefore, we have all higher order estimates of the second fundamental form at any point away from the singular set. By Theorem 2.6 and a diagonal sequence argument we can show that a subsequence of $\{(\Sigma_i^2, \mathbf{x}_i(t)), -1 < t < 1\}$ converges in smooth topology, possibly with multiplicities, to an embedded minimal surface Σ_∞ away from the singular set S_0 . Property (4)–(5) imply that the multiplicity of the convergence is bounded by some constant N_0 . The choice of N_0 and M_0 in (3.32) is clear from the above discussion. Since Σ_∞ is minimal and the convergence is smooth outside S_0 , the same argument as in Proposition 7.14 of Colding-Minicozzi [13] shows that $\Sigma_\infty \cup S_0$ is a smooth embedded minimal surface and the convergence is also in Hausdorff distance. The proposition is proved. $\square$

To study the multiplicity, we define a function

$$\Theta(x, r, t) := \lim_{i \rightarrow +\infty} \frac{\text{Area}_{g_i(t)}(\Sigma_{i,t} \cap B_r(x))}{\pi r^2}, \quad \forall (x, t) \in \Sigma_\infty \times (-1, 1). \quad (3.35)$$

Then the multiplicity at $(x, t) \in \Sigma_\infty \times (-1, 1)$ is give by

$$\mathfrak{m}(x, t) := \lim_{r \rightarrow 0} \Theta(x, r, t). \quad (3.36)$$

It is clear that $\mathfrak{m}(x, t)$ is an integer.

Lemma 3.14 *Under the assumption of Proposition 3.13, the function $\mathfrak{m}(x, t)$ is a constant integer on $\Sigma_\infty \times (-1, 1)$. Namely, $\mathfrak{m}(x, t)$ is independent of x and t .*

Proof We divide the proof into several steps.

Step 1. For each $t \in (-1, 1)$, $\mathfrak{m}(x, t)$ is constant on $\Sigma_\infty \setminus S_0$. Fix $t_0 \in (-1, 1)$ and $x_0 \in \Sigma_\infty \setminus S_0$. Since x_0 is a regular point, there exists $r_0 > 0$ such that for large i ,

$$|A|(x, t_0) \leq \frac{1}{r_0}, \quad \forall x \in B_{r_0}(x_0) \cap \Sigma_{i,t_0}. \quad (3.37)$$

By Lemma 3.2, we can assume r_0 small such that $B_{r_0}(x_0) \cap \Sigma_\infty$ can be written as a graph over the tangent plane of Σ_∞ at x_0 . Let $r_1 = \frac{r_0}{4}$. For any $p \in B_{r_1}(x_0) \cap \Sigma_{i,t_0}$, we have $B_{\frac{r_0}{2}}(p) \subset B_{r_0}(x_0)$. Thus, (3.37) implies that

$$|A|(x, t_0) \leq \frac{2}{r_0}, \quad \forall x \in B_{\frac{r_0}{2}}(p) \cap \Sigma_{i,t_0}. \quad (3.38)$$

By Lemma 3.3 for any $\delta > 0$ there exists $\rho_0 = \rho_0(r_0, \delta) \in (0, \frac{r_0}{400})$ such that for any $r \in (0, \rho_0)$ and any $p \in B_{r_1}(x_0) \cap \Sigma_{i,t_0}$ we have

$$\frac{\text{Area}_{g_i(t_0)}(C_p(B_r(p) \cap \Sigma_{i,t_0}))}{\pi r^2} \leq 1 + \delta. \quad (3.39)$$

On the other hand, by Lemma 3.5 we can choose ρ_0 small such that for any $r \in (0, \rho_0)$, on each component of $B_r(p) \cap \Sigma_{i,t_0}$ we obtain

$$\frac{\text{Area}_{g_i(t_0)}(C_p(B_r(p) \cap \Sigma_{i,t_0}))}{\pi r^2} \geq 1 - \delta. \quad (3.40)$$

Suppose that $B_{r_1}(x_0) \cap \Sigma_{i,t_0}$ has m_i connected components, where m_i is an integer bounded by a constant independent of i by Proposition 3.13. After taking a subsequence of $\{\Sigma_{i,t_0}\}$ if necessary, we can assume that m_i are the same integer denoted by m with $m \geq 1$. For any $x \in B_{\frac{r_1}{2}}(x_0) \cap \Sigma_\infty$, we denote by α_x the normal line passing through x of Σ_∞ . Since each component of $B_{r_1}(x_0) \cap \Sigma_{i,t_0}$ converges to $B_{r_1}(x_0) \cap \Sigma_\infty$ smoothly and $B_{r_1}(x_0) \cap \Sigma_\infty$ is a graph over the tangent plane of Σ_∞ at x_0 , α_x intersects transversally each component of Σ_{i,t_0} at exactly one point. Suppose that

$$\alpha_x \cap \left(B_{r_1}(x_0) \cap \Sigma_{i,t_0} \right) = \{p_i^{(1)}, p_i^{(2)}, \dots, p_i^{(m)}\}.$$

Then (3.39) and (3.40) imply that for any integer j with $1 \leq j \leq m$ and any $r \in (0, \rho_0)$,

$$1 - \delta \leq \frac{\text{Area}_{g_i(t_0)}(C_{p_i^{(j)}}(B_r(p_i^{(j)}) \cap \Sigma_{i,t_0}))}{\pi r^2} \leq 1 + \delta. \quad (3.41)$$

After shrinking r_0 if necessary, we can assume that $B_r(x) \cap \Sigma_\infty$ has only one component for any $r \in (0, \frac{r_1}{2})$ and any $x \in B_{\frac{r_1}{2}}(x_0) \cap \Sigma_\infty$. Since for any $1 \leq j \leq m$ we have $p_i^{(j)} \rightarrow x$ and $C_{p_i^{(j)}}(B_r(p_i^{(j)}) \cap \Sigma_{i,t_0})$ converges smoothly to $B_r(x) \cap \Sigma_\infty$, (3.41) implies that

$$m(1 - \delta) \leq \lim_{i \rightarrow +\infty} \frac{\text{Area}_{g_i(t_0)}(B_r(x) \cap \Sigma_{i,t_0})}{\pi r^2} \leq m(1 + \delta). \quad (3.42)$$

In other words, for any $x \in B_{\frac{r}{2}}(x_0) \cap \Sigma_\infty$ and any $r \in (0, \rho_0)$ we have

$$m(1 - \delta) \leq \Theta(x, r, t_0) \leq m(1 + \delta). \quad (3.43)$$

Taking $r \rightarrow 0$ in (3.43), we have

$$m(x, t_0) = m, \quad \forall x \in B_{\frac{r}{2}}(x_0) \cap \Sigma_\infty.$$

By the connectedness of $\Sigma_\infty \setminus \mathcal{S}_0$, we know that $m(x, t_0)$ is constant on $\Sigma_\infty \setminus \mathcal{S}_0$.

Step 2. For each $t \in (-1, 1)$, $m(x, t)$ is constant on Σ_∞ . It suffices to consider a singular point $p_0 \in \mathcal{S}_0$. Fix $t_0 \in (-1, 1)$. Suppose that $B_r(p_0) \cap \Sigma_\infty$ has no other singular points except p_0 for any $r \in (0, r_0)$. Then all points in $(B_r(p_0) \setminus B_\epsilon(p_0)) \cap \Sigma_\infty$ are regular and $(B_r(p_0) \setminus B_\epsilon(p_0)) \cap \Sigma_{i, t_0}$ has m connected components. Thus, we have

$$\begin{aligned} & \text{Area}_{g_i(t_0)}(\Sigma_{i, t_0} \cap B_r(p_0)) \\ & \leq \text{Area}_{g_i(t_0)}(\Sigma_{i, t_0} \cap (B_r(p_0) \setminus B_\epsilon(p_0))) + \text{Area}_{g_i(t_0)}(\Sigma_{i, t_0} \cap B_\epsilon(p_0)) \\ & \leq \text{Area}_{g_i(t_0)}(\Sigma_{i, t_0} \cap (B_r(p_0) \setminus B_\epsilon(p_0))) + N\epsilon^2, \end{aligned} \quad (3.44)$$

where we used (3.30) in the last inequality. Since each component of $\Sigma_{i, t_0} \cap (B_r(p_0) \setminus B_\epsilon(p_0))$ converges to $(B_r(p_0) \setminus B_\epsilon(p_0)) \cap \Sigma_\infty$ smoothly, we have

$$\lim_{i \rightarrow +\infty} \text{Area}_{g_i(t_0)}(\Sigma_{i, t_0} \cap (B_r(p_0) \setminus B_\epsilon(p_0))) = m \text{Area}_{g_\infty}(\Sigma_\infty \cap (B_r(p_0) \setminus B_\epsilon(p_0))), \quad (3.45)$$

where m is the number of components of $\Sigma_{i, t_0} \cap (B_r(p_0) \setminus B_\epsilon(p_0))$. Note that m is also the multiplicity at each regular point in Σ_∞ by Step 1. Combining (3.44) with (3.45), we have

$$\begin{aligned} & m \text{Area}_{g_\infty}(\Sigma_\infty \cap (B_r(p_0) \setminus B_\epsilon(p_0))) \\ & \leq \lim_{i \rightarrow +\infty} \text{Area}_{g_i(t_0)}(\Sigma_{i, t_0} \cap B_r(p_0)) \\ & \leq m \text{Area}_{g_\infty}(\Sigma_\infty \cap (B_r(p_0) \setminus B_\epsilon(p_0))) + N\epsilon^2. \end{aligned} \quad (3.46)$$

Taking $\epsilon \rightarrow 0$ in (3.46), we have

$$\lim_{i \rightarrow +\infty} \text{Area}_{g_i(t_0)}(\Sigma_{i, t_0} \cap B_r(p_0)) = m \text{Area}_{g_\infty}(\Sigma_\infty \cap B_r(p_0)). \quad (3.47)$$

Thus, we have

$$\begin{aligned} m(p_0, t_0) &= \lim_{r \rightarrow 0} \frac{\text{Area}_{g_i(t_0)}(\Sigma_{i, t_0} \cap B_r(p_0))}{\pi r^2} \\ &= m \lim_{r \rightarrow 0} \frac{\text{Area}_{g_\infty}(\Sigma_\infty \cap B_r(p_0))}{\pi r^2} = m. \end{aligned}$$

This implies that the multiplicity of each singular point is the same as that of any regular point.

Step 3. $m(x, t)$ is constant in t . Let $x_0 \in \Sigma_\infty$ and $\epsilon_i := \max_{\Sigma_{i,t}} |H| \rightarrow 0$. Similar to the proof of (3.5), for any $t_1, t_2 \in (-1, 1)$ we have

$$|x_i(p, t_2) - x_0| \leq |x_i(p, t_1) - x_0| + \epsilon_i |t_1 - t_2|. \quad (3.48)$$

Thus, we have

$$x(t_1)^{-1}(B_r(x_0) \cap \Sigma_{i,t_1}) \subset x(t_2)^{-1}(B_{r_1}(x_0) \cap \Sigma_{i,t_2}), \quad (3.49)$$

where

$$r_1 = r + \epsilon_i |t_2 - t_1|. \quad (3.50)$$

Recall that the evolution of area element along mean curvature flow is dominated by $|H|^2$ as in (3.3). Therefore (3.49) implies that

$$\text{Area}_{g_i(t_1)}(B_r(x_0) \cap \Sigma_{i,t_1}) \leq e^{\epsilon_i^2 |t_2 - t_1|} \text{Area}_{g_i(t_2)}(B_{r_1}(x_0) \cap \Sigma_{i,t_2}). \quad (3.51)$$

Combining (3.51) with (3.50), we have

$$\begin{aligned} \Theta(x_0, r, t_1) &= \lim_{i \rightarrow +\infty} \frac{\text{Area}_{g_i(t_1)}(B_r(x_0) \cap \Sigma_{i,t_1})}{\pi r^2} \\ &\leq \lim_{i \rightarrow +\infty} e^{\epsilon_i^2 |t_2 - t_1|} \frac{\text{Area}_{g_i(t_2)}(B_{r_1}(x_0) \cap \Sigma_{i,t_2})}{\pi r^2} \\ &= \lim_{i \rightarrow +\infty} \frac{\text{Area}_{g_i(t_2)}(B_{r_1}(x_0) \cap \Sigma_{i,t_2})}{\pi r^2} \\ &= \Theta(x_0, r, t_2). \end{aligned} \quad (3.52)$$

Letting $r \rightarrow 0$ in (3.52), we have

$$m(x_0, t_1) \leq m(x_0, t_2).$$

Since t_1 and t_2 are arbitrary in $(-1, 1)$, we have that $m(x, t)$ is constant in t . $\square$

4 Multiplicity-one convergence of the rescaled mean curvature flow

In this section, we show that a rescaled mean curvature flow with mean curvature exponential decay will converge smoothly to a plane with multiplicity one.

Theorem 4.1 *Let $\{(\Sigma^2, \mathbf{x}(t)), 0 \leq t < +\infty\}$ be a rescaled mean curvature flow*

$$\left(\frac{\partial \mathbf{x}}{\partial t}\right)^\perp = -\left(H - \frac{1}{2}\langle \mathbf{x}, \mathbf{n} \rangle\right) \mathbf{n} \quad (4.1)$$

satisfying

$$d(\Sigma_t, 0) \leq D, \quad \text{and} \quad \max_{\Sigma_t} |H(p, t)| \leq \Lambda_0 e^{-\frac{t}{2}} \quad (4.2)$$

for two constants $D, \Lambda_0 > 0$. Then there exists a sequence of times $t_j \rightarrow +\infty$ such that Σ_{t_j} converge in smooth topology to a plane passing through the origin with multiplicity one.

We sketch the proof of Theorem 4.1. First, we show the weak compactness for any sequence of the rescaled mean curvature flow in Lemma 4.2. Suppose that the multiplicity is at least two. By using the decomposition of spaces (c.f. Definition 4.4) we can select a special sequence $\{t_i\}$ in Lemma 4.8. This special sequence is needed to control the upper bound of the function w_i by using the parabolic Harnack inequality (c.f. Lemma 4.14). Then we can take the limit for the function w_i and obtain a positive function w with uniform bounds (c.f. Proposition 4.9). The function w satisfies the linearized rescaled mean curvature flow equation. The bounds of w imply the L -stability of the limit plane (c.f. Lemmas 4.16 and 4.17). However, the plane is not L -stable and we obtain a contradiction.

4.1 Convergence away from singularities

Lemma 4.2 *Under the assumption of Theorem 4.1, for any sequence $t_i \rightarrow +\infty$, there is a plane Σ_∞ passing through the origin and a finite set $\mathcal{S}_0 \subset \Sigma_\infty$ of points satisfying the following properties. For any $T > 0$, there is a subsequence, still denoted by $\{t_i\}$, such that $\{\Sigma_{t_i+t}, -T < t < T\}$ converges in smooth topology, possibly with multiplicities at most N_0 , to the plane Σ_∞ away from the space-time singular set $\mathcal{S} = \{(x, t) \mid t \in (-T, T), x \in e^{\frac{t}{2}} \mathcal{S}_0\}$.*

Proof The proof divides into the following steps.

Step 1. The energy of Σ_t is uniformly bounded along the flow (4.1). In fact, we rescale the flow Σ_t by

$$s = 1 - e^{-t}, \quad \hat{\Sigma}_s = \sqrt{1-s} \Sigma_{-\log(1-s)} \quad (4.3)$$

such that $\{\hat{\Sigma}_s, 0 \leq s < 1\}$ is a mean curvature flow satisfying (2.1). Moreover, the mean curvature $\hat{H}$ of $\hat{\Sigma}_s$ satisfies

$$\max_{\hat{\Sigma}_s} |\hat{H}| = \frac{1}{\sqrt{1-s}} \max_{\Sigma_t} |H| = e^{\frac{t}{2}} \max_{\Sigma_t} |H| \leq \Lambda_0, \quad (4.4)$$

where we used the assumption (4.2). Note that the scalar curvature $\hat{S}$ of $\hat{\Sigma}_s$ satisfies $\hat{S} = \hat{H}^2 - |\hat{A}|^2$, where $\hat{A}$ denotes the second fundamental form of $\hat{\Sigma}_s$. By the Gauss-Bonnet theorem we have

$$\int_{\hat{\Sigma}_s} |\hat{A}|^2 d\hat{\mu}_s = \int_{\hat{\Sigma}_s} |\hat{H}|^2 d\hat{\mu}_s - 4\pi \chi(\Sigma_0) \leq \Lambda_0^2 \text{Area}(\Sigma_0) - 4\pi \chi(\Sigma_0) =: \Lambda. \quad (4.5)$$

where we used the fact that $\text{Area}(\hat{\Sigma}_s)$ is non-increasing in s . Here $\chi(\Sigma)$ denotes the Euler characteristic of Σ . Therefore, by Lemma 2.3 the energy of Σ_t satisfies the inequality

$$\int_{\Sigma_t} |A|^2 d\mu_t \leq \Lambda. \quad (4.6)$$

Step 2. For any sequence $t_i \rightarrow +\infty$, we can obtain a refined sequence converging to a limit minimal surface $\tilde{\Sigma}_\infty$. For any sequence $t_i \rightarrow +\infty$, we can rescale the flow Σ_t by

$$s = 1 - e^{-(t-t_i)}, \quad \tilde{\Sigma}_{i,s} = \sqrt{1-s} \Sigma_{t_i - \log(1-s)} \quad (4.7)$$

such that for each i the flow $\{\tilde{\Sigma}_{i,s}, 1 - e^{t_i} \leq s < 1\}$ is a mean curvature flow satisfying (2.1) with the following properties:

(a). For any small $\lambda > 0$, the mean curvature of $\tilde{\Sigma}_{i,s}$ satisfies

$$\lim_{i \rightarrow +\infty} \max_{\tilde{\Sigma}_{i,s} \times [1 - e^{t_i}, 1 - \lambda]} |\tilde{H}_i|(p, s) = 0;$$

(b). The energy of $\tilde{\Sigma}_{i,s}$ satisfies (4.6);

(c). Uniform upper bound on the area ratio;

(d). Uniform lower bound on the area ratio;

(e). There exists a constant $D' > 0$ such that $d(\tilde{\Sigma}_{i,s}, 0) \leq D'$ for any i .

In fact, Property (a) and (e) follow from the assumption (4.2), and Property (b) follows from (4.6). Property (c) follows from Lemma 2.1 and Lemma 2.3, and Property (d) follows directly from Lemma 3.5. Therefore, by Definition 3.12 for any $T_0 > 2$, small $\lambda \in (0, 1)$ and any $s_0 \in [-T_0 + 1, -\lambda]$ the sequence $\{\tilde{\Sigma}_{i,s_0+\tau}, -1 < \tau < 1\}$ is a refined sequence. By Proposition 3.13 a subsequence of $\{\tilde{\Sigma}_{i,s_0+\tau}, -1 < \tau < 1\}$ converges in smooth topology, possibly with multiplicity at most N_0 , to a smooth embedded minimal surface $\tilde{\Sigma}_\infty$ away from a finite set of points $\tilde{\mathcal{S}} = \{q_1, \dots, q_l\}$ such that

$$l \leq \left\lceil \frac{\Lambda}{\epsilon_0} \right\rceil + 1 = M_0, \quad (4.8)$$

where we used (3.32) and (4.6).

Claim 4.3 *Under the above assumptions, there exists a subsequence of $\{\tilde{\Sigma}_{i,s}, -T_0 < s < 1 - \lambda\}$ such that it converges in smooth topology, possibly with multiplicity at most N_0 , to the limit minimal surface $\tilde{\Sigma}_\infty$ away from the singular set $\tilde{S} = \{q_1, \dots, q_l\}$. Furthermore, $\tilde{\Sigma}_\infty$ and $\tilde{S}$ are independent of T_0 and λ .*

Proof Let $T_0 = N + \beta$ where $N \in \mathbb{N}, N \geq 2$ and $\beta \in [0, 1)$. For the interval $I_0 := (-2, 0)$, we have a subsequence of $\{t_i\}$, which we denote by $\{i_k^{(1)}\}$, such that $\{\tilde{\Sigma}_{i_k^{(1)},s}, s \in I_0\}$ converges in smooth topology, possibly with multiplicity at most N_0 , to a limit minimal surface $\tilde{\Sigma}_\infty$ away from a singular set $\tilde{S} = \{q_1, \dots, q_l\}$. Consider the interval $I'_0 := (-1 - \lambda, 1 - \lambda)$ with $\lambda \in (0, 1)$. Since $I'_0 \cap I_0 \neq \emptyset$, we can take a further subsequence of $\{i_k^{(1)}\}$, denoted by $\{i_k^{(2)}\}$, such that $\{\tilde{\Sigma}_{i_k^{(2)},s}, s \in I'_0\}$ converges to the same limit surface $\tilde{\Sigma}_\infty$ away from the same singular set $\tilde{S}$. Similarly, we consider $I_2 = (-2 - \beta, -\beta)$. Since $I_0 \cap I_2 \neq \emptyset$, we can take a subsequence of $\{i_k^{(2)}\}$, denoted by $\{i_k^{(3)}\}$, such that $\{\tilde{\Sigma}_{i_k^{(3)},s}, s \in I_2\}$ converges to the same limit surface $\tilde{\Sigma}_\infty$ away from the same singular set $\tilde{S}$. Repeating this process for the interval $I_j = (-j - \beta, -j - \beta + 2)$ for $j = 3, \dots, N$ and we get a subsequence $\{i_k^{(N+1)}\}$ of $\{t_i\}$, such that $\{\tilde{\Sigma}_{i_k^{(N+1)},s}, -T_0 < s < 1 - \lambda\}$ converges to $\tilde{\Sigma}_\infty$ away from $\tilde{S}$. By the construction of $\tilde{\Sigma}_\infty$ and $\tilde{S}$, we know that $\tilde{\Sigma}_\infty$ and $\tilde{S}$ are independent of the choice of T_0 and λ . The Claim is proved. $\square$

Step 3. Each limit $\tilde{\Sigma}_\infty$ must be a plane through the origin. In fact, by Huisken's monotonicity formula, along the rescaled mean curvature flow (4.1) we have

$$\frac{d}{dt} \int_{\Sigma_t} e^{-\frac{|\mathbf{x}|^2}{4}} d\mu_t = - \int_{\Sigma_t} e^{-\frac{|\mathbf{x}|^2}{4}} \left| H - \frac{1}{2} \langle \mathbf{x}, \mathbf{n} \rangle \right|^2 d\mu_t. \quad (4.9)$$

This implies that

$$\int_0^\infty \int_{\Sigma_t} e^{-\frac{|\mathbf{x}|^2}{4}} \left| H - \frac{1}{2} \langle \mathbf{x}, \mathbf{n} \rangle \right|^2 d\mu_t < +\infty.$$

For any $t_i \rightarrow +\infty$, we rescale the flow (4.1) by (4.7) such that for each i the flow $\{\tilde{\Sigma}_{i,s}, 1 - e^{t_i} \leq s < 1\}$ is a mean curvature flow satisfying (2.1) and we denote the solution by $\tilde{\mathbf{x}}_{i,s}$. Therefore, for fixed $T_0 > 0$, small $\lambda > 0$ and large i we have

$$\begin{aligned}
& \lim_{i \rightarrow +\infty} \int_{-T_0}^{1-\lambda} ds \int_{\tilde{\Sigma}_{i,s}} e^{-\frac{|\tilde{\mathbf{x}}_{i,s}|^2}{4(1-s)}} \left| \tilde{H}_i - \frac{1}{2(1-s)} \langle \tilde{\mathbf{x}}_{i,s}, \mathbf{n} \rangle \right|^2 d\tilde{\mu}_{i,s} \\
&= \lim_{t_i \rightarrow +\infty} \int_{t_i - \log(1+T_0)}^{t_i - \log \lambda} \int_{\Sigma_t} e^{-\frac{|\mathbf{x}|^2}{4}} \left| H - \frac{1}{2} \langle \mathbf{x}, \mathbf{n} \rangle \right|^2 d\mu_t = 0.
\end{aligned}$$

Since $\{\tilde{\Sigma}_{i,s}, -T_0 < s < 1 - \lambda\}$ converges locally smoothly, possibly with multiplicity at most N_0 , to $\tilde{\Sigma}_\infty$ away from $\tilde{\mathcal{S}}$, we have

$$\int_{-T_0}^{1-\lambda} ds \int_{\tilde{\Sigma}_\infty} e^{-\frac{|\tilde{\mathbf{x}}_\infty|^2}{4(1-s)}} \left| \tilde{H} - \frac{1}{2(1-s)} \langle \tilde{\mathbf{x}}_\infty, \mathbf{n} \rangle \right|^2 d\tilde{\mu}_{\infty,s} = 0.$$

Therefore, $\{(\tilde{\Sigma}_\infty, \tilde{\mathbf{x}}_\infty(p, s)), -T_0 < s < 1 - \lambda\}$ satisfies the equation

$$\tilde{H} - \frac{1}{2(1-s)} \langle \tilde{\mathbf{x}}_\infty, \mathbf{n} \rangle = 0$$

away from the singular set $\tilde{\mathcal{S}}$. Since $\tilde{\Sigma}_\infty$ is a smooth embedded minimal surface, we have $\tilde{H} = \langle \tilde{\mathbf{x}}_\infty, \mathbf{n} \rangle = 0$. By Lemma 2.2 we know $\tilde{\Sigma}_\infty$ must be a plane passing through the origin. Let $\mathbf{x}_i(p, t) = \mathbf{x}(p, t_i + t)$ and $\Sigma_{i,t} = \Sigma_{t_i+t}$. Since $\{\tilde{\Sigma}_{i,s}, -T_0 < s < 1 - \lambda\}$ converges locally smoothly to $\tilde{\Sigma}_\infty$ away from $\tilde{\mathcal{S}}$, the flow $\{\Sigma_{i,t}, -\log(1+T_0) < t < -\log \lambda\}$ also converges locally smoothly to the plane $\tilde{\Sigma}_\infty$ away from the singular set $\mathcal{S} = \{(x, t) \mid t \in (-\log(1+T_0), -\log \lambda), x \in e^{\frac{t}{2}} \tilde{\mathcal{S}}\}$. Moreover, by Claim 4.3 $\tilde{\Sigma}_\infty$ and $\tilde{\mathcal{S}}$ are independent of T_0 and λ . The lemma is proved. $\square$

4.2 Decomposition of spaces and a “monotone decreasing” quantity

Note that if one limit plane in Lemma 4.2 has multiplicity one, then the proof of Theorem 4.1 is done. Therefore, we can assume that every limit plane has multiplicity more than one. Consequently, we can decompose the space as follows.

Definition 4.4(1). We define the set $\mathbf{S} = \mathbf{S}(\epsilon, \Sigma_t) = \{y \in \Sigma_t \mid |y| < \epsilon^{-1}, |A|(y, t) > \epsilon^{-1}\}$.

(2). The ball $B_{\epsilon^{-1}}(0)$ can be decomposed into three parts as follows:

- the high curvature part $\mathbf{H}$, which is defined by $\mathbf{H} = \mathbf{H}(\epsilon, \Sigma_t) = \{x \in \mathbb{R}^3 \mid |x| < \epsilon^{-1}, d(x, \mathbf{S}) < \frac{\epsilon}{2}\}$. Here d denotes the Euclidean distance in $\mathbb{R}^3$.
- the thick part $\mathbf{TK}$, which is defined by

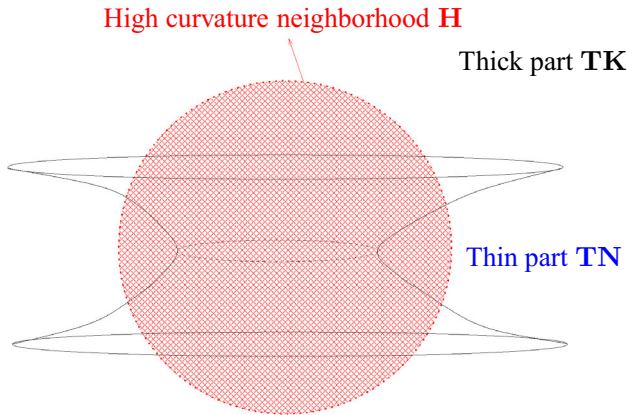


Fig. 1 Decomposition of the space

$$\begin{aligned}
 \mathbf{TK} &= \mathbf{TK}(\epsilon, \Sigma_t) \\
 &= \left\{ x \in \mathbb{R}^3 \mid |x| < \epsilon^{-1}, \right. \\
 &\quad \text{there is a continuous curve } \gamma \subset B_{\epsilon^{-1}}(0) \setminus (\mathbf{H} \cup \Sigma_t) \\
 &\quad \left. \text{connecting } x \text{ and some } y \text{ with } B(y, \epsilon) \subset B_{\epsilon^{-1}}(0) \setminus (\mathbf{H} \cup \Sigma_t) \right\}.
 \end{aligned}$$

- the thin part $\mathbf{TN}$, which is defined by $\mathbf{TN} = \mathbf{TN}(\epsilon, \Sigma_t) = B_{\epsilon^{-1}}(0) \setminus (\mathbf{H} \cup \mathbf{TK})$.

We remind the readers that the decomposition in the above definition depends on the fact that the limit plane has multiplicity more than one. Intuitively, the high curvature part $\mathbf{H}$ is the neighborhood of points with large second fundamental form (c.f. Fig. 1). The thin part $\mathbf{TN}$ is the domain between the top and bottom sheets. The thick part is the union of path connected components of the domain “outside” the sheets. Note that in the above definition, the existence of curve γ is to guarantee the path-connectedness. Because of the boundary issue, some points in $\mathbf{TK}$ may be very close to the points in $\mathbf{TN}$ or $\mathbf{H}$.

Remark 4.5 The decomposition of space is motivated by the decomposition of Ricci flow time slices in Chen-Wang [8].

Lemma 4.6 *For each fixed ϵ , we have $\lim_{t \rightarrow \infty} |\mathbf{TN}(\epsilon, \Sigma_t)| = 0$. Here the notation $|\Omega|$ denotes the volume of Ω with respect to the standard metric on $\mathbb{R}^3$.*

Proof For otherwise, we can find an ϵ and a sequence $t_i \rightarrow \infty$ such that

$$\lim_{i \rightarrow \infty} |\mathbf{TN}(\epsilon, \Sigma_{t_i})| \geq \kappa_0 > 0. \quad (4.10)$$

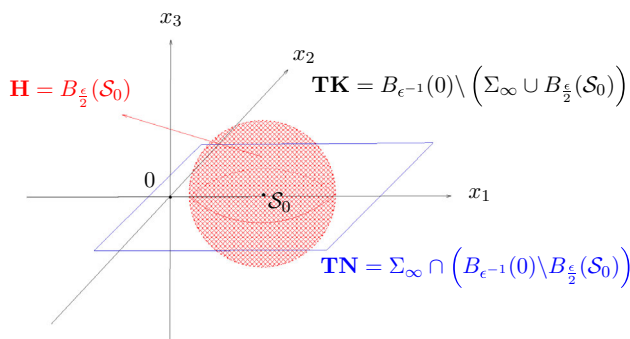


Fig. 2 Decomposition in the limit space

However, Σ_{t_i} converges locally smoothly to a plane Σ_∞ passing through the origin away from the singular set $S_0 \subset \Sigma_\infty$. Therefore, the sets $S(\epsilon, \Sigma_{t_i})$ converge to S_0 as $i \rightarrow +\infty$. Without loss of generality, we can assume that Σ_∞ is determined by $x_3 = 0$ and the singular set $S_0 = \{(1, 0, 0)\}$. Then the high curvature parts $H(\epsilon, \Sigma_{t_i})$ converge to $B_{\frac{\epsilon}{2}}((1, 0, 0))$ (c.f. Fig. 2).

Note that $B_{\epsilon^{-1}}(0) \setminus (\Sigma_\infty \cup B_{\frac{\epsilon}{2}}((1, 0, 0)))$ contains two parts $\hat{B}^+ \cup \hat{B}^-$, where

$$\begin{aligned} \hat{B}^+ &= \left\{ (x_1, x_2, x_3) \mid x_1^2 + x_2^2 + x_3^2 < \epsilon^{-2}, (x_1 - 1)^2 + x_2^2 + x_3^2 > \frac{\epsilon^2}{4}, x_3 > 0 \right\}, \\ \hat{B}^- &= \left\{ (x_1, x_2, x_3) \mid x_1^2 + x_2^2 + x_3^2 < \epsilon^{-2}, (x_1 - 1)^2 + x_2^2 + x_3^2 > \frac{\epsilon^2}{4}, x_3 < 0 \right\}. \end{aligned}$$

Fix an arbitrary point $x \in \hat{B}^+$. It is not hard to see that x can be connected to $(0, 0, 1)$ by a continuous curve $\gamma \subset \hat{B}^+$. Furthermore, it is clear that $B_\epsilon((0, 0, 1)) \cap B_\epsilon((1, 0, 0)) = \emptyset$. Similar argument applies for $\hat{B}^-$. Therefore, we see that $\hat{B}^+ \cup \hat{B}^-$ are contained in the limit of $\mathbf{TK}(\epsilon, \Sigma_{t_i})$. Recall that $H(\epsilon, \Sigma_{t_i})$ converges to $B_{\frac{\epsilon}{2}}((1, 0, 0))$. Following the definition of $\mathbf{TN}$, the Hausdorff limit of $\mathbf{TN}(\epsilon, \Sigma_{t_i})$ is a subset of $(B_{\epsilon^{-1}}(0) \cap \Sigma_\infty) \setminus B_{\frac{\epsilon}{2}}((1, 0, 0))$. Thus, for large t_i we know $\mathbf{TN}(\epsilon, \Sigma_{t_i})$ lies in a small neighborhood of $(B_{\epsilon^{-1}}(0) \cap \Sigma_\infty) \setminus B_{\frac{\epsilon}{2}}((1, 0, 0))$. In other words, for any $\delta > 0$ and large t_i we have $\mathbf{TN}(\epsilon, \Sigma_{t_i}) \subset \Omega_{\epsilon, \delta}$, where

$$\Omega_{\epsilon, \delta} = \left\{ (x_1, x_2, x_3) \in B_{\epsilon^{-1}}(0) \mid (x_1 - 1)^2 + x_2^2 + x_3^2 \geq \frac{\epsilon^2}{16}, |x_3| < \delta \right\}.$$

Then for large t_i ,

$$|\mathbf{TN}(\epsilon, \Sigma_{t_i})| \leq |\Omega_{\epsilon, \delta}|. \quad (4.11)$$

Taking $i \rightarrow \infty$ in (4.11) we have that the limit of $|\mathbf{TN}(\epsilon, \Sigma_{t_i})|$ is arbitrary small. This contradicts the assumption (4.10). $\square$

It is not hard to observe that

Lemma 4.7 *If Σ_∞ has multiplicity more than one, then for sufficiently large t_i we have*

$$|\mathbf{TN}(\epsilon, \Sigma_{t_i})| > 0.$$

Proof Since Σ_t is embedded and Σ_{t_i} converges locally smoothly to the limit plane Σ_∞ , all components of $(\Sigma_{t_i} \cap B_{\epsilon^{-1}}(0)) \setminus \mathbf{H}(\epsilon, \Sigma_{t_i})$ lie in the $\frac{\epsilon}{2}$ -neighborhood of the plane Σ_∞ for large t_i . If Σ_∞ has multiplicity more than one, the space between the top and bottom sheets of $(\Sigma_{t_i} \cap B_{\epsilon^{-1}}(0)) \setminus \mathbf{H}(\epsilon, \Sigma_{t_i})$ belongs to $\mathbf{TN}$. Thus, by the definition of $\mathbf{TN}$ we have that $\mathbf{TN}(\epsilon, \Sigma_{t_i})$ is nonempty and

$$|\mathbf{TN}(\epsilon, \Sigma_{t_i})| > 0.$$

□

However, in general we do not know whether $|\mathbf{TN}(\epsilon, \Sigma_t)|$ is a continuous function of t , but we can choose t_i carefully such that $|\mathbf{TN}(\epsilon, \Sigma_t)|$ is bounded on a time interval. Combining Lemma 4.6 with Lemma 4.7, we have

Lemma 4.8 *There is a sequence of times $t_i \rightarrow \infty$ such that*

$$\sup_{t_i \leq t \leq t_i + 1} |\mathbf{TN}(\epsilon, \Sigma_t)| \leq 2|\mathbf{TN}(\epsilon, \Sigma_{t_i})|. \quad (4.12)$$

Proof By Lemma 4.7 we can find $s_1 \geq 1$ such that

$$|\mathbf{TN}(\epsilon, \Sigma_{s_1})| > 0. \quad (4.13)$$

We search for time $t \in [s_1, s_1 + 1]$ satisfying $|\mathbf{TN}(\epsilon, \Sigma_t)| > 2|\mathbf{TN}(\epsilon, \Sigma_{s_1})|$. If no such time exists, then we set $t_1 = s_1$. Otherwise, we choose such a time and denote it by $s_1^{(1)}$. Then we search the time interval $[s_1^{(1)}, s_1^{(1)} + 1]$. Inductively, we search $[s_1^{(k)}, s_1^{(k)} + 1]$. If we have

$$\sup_{t \in [s_1^{(k)}, s_1^{(k)} + 1]} |\mathbf{TN}(\epsilon, \Sigma_t)| \leq 2|\mathbf{TN}(\epsilon, \Sigma_{s_1^{(k)}})|,$$

then we denote $t_1 = s_1^{(k)}$ and stop the searching process. Otherwise, we choose a time $s_1^{(k+1)} \in [s_1^{(k)}, s_1^{(k)} + 1]$ with more than doubled $|\mathbf{TN}|$ value and continue the process. Note that

$$|\mathbf{TN}(\epsilon, \Sigma_{s_1^{(k)}})| \geq 2^k |\mathbf{TN}(\epsilon, \Sigma_{s_1})| \rightarrow \infty, \quad \text{as } k \rightarrow \infty,$$

where we used (4.13). Since by Lemma 4.6 $|\mathbf{TN}(\epsilon, \Sigma_t)|$ tends to zero, this process must stop in finite steps. After we find t_1 , set $s_2^{(0)} = t_1 + 1$ and continue the previous process to find time in $[s_2^{(0)}, s_2^{(0)} + 2]$ with more than doubled $|\mathbf{TN}|$ -value, with slight change that the time-interval has length 2. Similarly, for some finite k , we have

$$\sup_{t \in [s_2^{(k)}, s_2^{(k)} + 2]} |\mathbf{TN}(\epsilon, \Sigma_t)| \leq 2|\mathbf{TN}(\epsilon, \Sigma_{s_2^{(k)}})|.$$

Then we define $t_2 = s_2^{(k)}$. Inductively, after we find t_l , we set $s_{l+1}^{(0)} = t_l + l$. Then we start the process to search time in $[s_{l+1}^{(0)}, s_{l+1}^{(0)} + l + 1]$ with more than doubled $|\mathbf{TN}|$ value. This process is well defined. From its construction, it is clear that $t_i \rightarrow \infty$ and satisfies (4.12). The lemma is proved. $\square$

Let $O(3)$ be the group of all rotations in $\mathbb{R}^3$. It is clear that $O(3)$ is compact and preserves many geometric quantities including distance, mean curvature and second fundamental form. Suppose $\mathbf{x}(t)$ is a rescaled mean curvature flow solution of (4.1), then $\sigma \circ \mathbf{x}$ is again a solution of (4.1) for each fixed $\sigma \in O(3)$. Therefore, up to rotations, we can always assume the limit plane in Lemma 4.2 to be a horizontal plane. For if $\Sigma_{i,t}$ converges to some plane P' passing through the origin, we can choose a rotation σ such that $\sigma(P')$ is a horizontal plane. Then it is clear that $\sigma \circ \Sigma_{i,t}$ converges to the horizontal plane $\sigma(P')$. It is also important to note that the quantities defined in Definition 4.4 are invariant under the action of $\sigma \in O(3)$. Consequently, for the flow $\sigma \circ \Sigma_{i,t}$, the estimate (4.12) becomes

$$\sup_{t_i \leq t \leq t_i + i} |\mathbf{TN}(\epsilon, \sigma \circ \Sigma_t)| \leq 2|\mathbf{TN}(\epsilon, \sigma \circ \Sigma_{t_i})|.$$

Therefore, replacing $\Sigma_{i,t}$ by $\sigma \circ \Sigma_{i,t}$ if necessary, we can always assume the limit plane is horizontal and (4.12) holds.

4.3 Construction of auxiliary functions

Let P be the horizontal plane

$$P := \{(x_1, x_2, x_3) \mid x_3 = 0\} \subset \mathbb{R}^3. \quad (4.14)$$

For each pair of fixed positive numbers r_1, r_2 with $r_1 < r_2$, we define

$$\text{Ann}(r_1, r_2) := \left(B_{r_2}(0) \setminus B_{r_1}(0) \right) \cap P. \quad (4.15)$$

Based on the choice of ϵ and subsequences in Lemma 4.8, on $\text{Ann}(\epsilon, \epsilon^{-1}) \times [0, \infty)$, we shall construct an auxiliary smooth function w which will be used to show the L -stability of the plane P in the next subsection (c.f. Lemma 4.16 and Lemma 4.17). The existence of such w , which is the main result of this subsection, is provided by the the following proposition.

Proposition 4.9 *Under the assumption of Theorem 4.1, if the multiplicity of the convergence is at least two, then for each sufficiently small $\epsilon > 0$, we can find a smooth function w defined on $\text{Ann}(\epsilon, \epsilon^{-1}) \times [0, \infty)$ such that*

$$\frac{\partial w}{\partial t} = Lw := \Delta_0 w - \frac{1}{2}\langle x, \nabla w \rangle + \frac{1}{2}w, \quad (4.16)$$

where Δ_0 is the standard Laplacian operator on the plane P . Furthermore, there is a constant $C > 0$ independent of t such that

$$0 < w(x, t) < C, \quad \forall (x, t) \in \text{Ann}(\epsilon, \epsilon^{-1}) \times [0, \infty), \quad (4.17)$$

$$w(x, 0) > \frac{1}{C}, \quad \forall x \in \text{Ann}(\epsilon, \epsilon^{-1}). \quad (4.18)$$

The function w in (4.16) is the renormalized “height-difference” function. Its existence boils down to the discussion in Sects. 4.1 and 4.2. In summary, before we prove Proposition 4.9, we already have the following properties in the remainder discussion in this subsection.

- (1). ϵ is fixed.
- (2). The sequence $\{t_i\}$ is chosen as in Lemma 4.8. Thus, the volume of $\mathbf{TN}$ increases slowly in the sense that

$$\sup_{t_i \leq t \leq t_i + i} |\mathbf{TN}(\epsilon, \Sigma_t)| \leq 2|\mathbf{TN}(\epsilon, \Sigma_{t_i})|. \quad (4.19)$$

- (3). For any $T > 0$, there is a subsequence, still denoted by $\{t_i\}$, such that $\{\Sigma_{i,t}, -T < t < T\}$ converges to a limit flow, with multiplicity N_0 , on the plane

$$\Sigma_\infty = P = \{(x_1, x_2, x_3) \mid x_3 = 0\} \subset \mathbb{R}^3. \quad (4.20)$$

Here we assumed $\Sigma_{i,t} = \Sigma_{t_i+t}$ as before. The convergence is smooth away from $\mathcal{S} = \{(x, t) \mid x \in \mathcal{S}_t, t \in (-T, T)\}$, which satisfies

$$\mathcal{S}_t = e^{\frac{t}{2}} \mathcal{S}_0, \quad (4.21)$$

The set $\mathcal{S}_0 = \{p_1, p_2, \dots, p_l\}$ satisfies

$$l \leq M_0, \quad (4.22)$$

where M_0 is the constant in the right-hand side of (4.8).

- (4). By Lemma 4.2, the limit plane P and S_0 are independent of the choice of T . In other words, for different T we have different subsequences of $\{\Sigma_{i,t}, -T < t < T\}$ but these two subsequences converge to the same limit plane P with the same S_0 .

The following lemma is a direct consequence of (4.21).

Lemma 4.10 *There exists $t_\epsilon > 0$ depending only on ϵ and S_0 such that*

$$\text{Ann}(\epsilon, \epsilon^{-1}) \cap S_t = \emptyset, \quad \forall t > t_\epsilon. \quad (4.23)$$

Furthermore, if $0 \notin S_0$, then (4.23) can be improved as

$$(B_{\epsilon^{-1}}(0) \cap P) \cap S_t = \emptyset, \quad \forall t > t_\epsilon. \quad (4.24)$$

Proof In light of (4.21), all points in $S_t \setminus \{0\}$ will move further away from the point $\{0\}$ when t is increasing. Set

$$\min S_0 := \min\{|y| \mid y \in S_0 \setminus \{0\}\} > 0, \quad t_\epsilon := 2 \log \frac{1}{\epsilon \min S_0}. \quad (4.25)$$

Then for any $t \geq t_\epsilon$ and any $y_0 \in S_0 \setminus \{0\}$ we have $e^{\frac{t}{2}}|y_0| \geq \epsilon^{-1}$. This implies that

$$(S_t \setminus \{0\}) \cap B_{\epsilon^{-1}}(0) = \emptyset, \quad \forall t > t_\epsilon. \quad (4.26)$$

Then both (4.23) and (4.24) follow from (4.26). $\square$

For the convergence from $\Sigma_{i,t}$ to Σ_∞ , it is clear that the origin 0 is a natural base point. It is possible that $0 \in S_0$. If this is the case, then $0 \in S_t$ for each t , which is very different from all other singular points in S_0 by Lemma 4.10. Therefore, if $0 \in S_0$, it is essential in the sense that it cannot be perturbed away by varying t . However, for later application, we prefer to choose a base point uniformly away from singular set, whose existence is guaranteed by the following lemma.

Lemma 4.11 *There exists a point $x_0 \in \text{Ann}(2\epsilon, 1)$ such that*

$$(B_{4\epsilon}(x_0) \cap \Sigma_\infty) \cap S_t = \emptyset, \quad \forall t \in (-\infty, \infty). \quad (4.27)$$

Proof Denote by $\gamma_j = \{sp_j \mid s \geq 0\}$, where p_j is any point in $S_0 = \{p_1, p_2, \dots, p_l\}$. It is clear that γ_j is the ray passing through p_j with the initial point 0 if $p_j \neq 0$ and $\gamma_j = \{0\}$ if $p_j = 0$. Recall that the number of

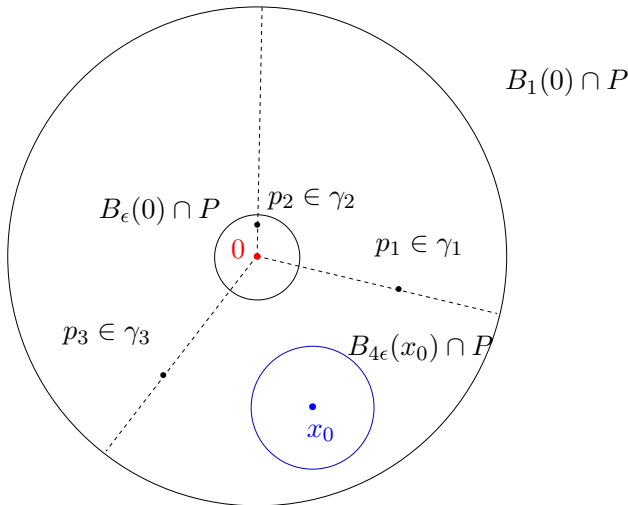


Fig. 3 An alternate base point of the limit plane

points in $\mathcal{S}_0$ is uniformly bounded by (4.8) and ϵ is very small, we can find a point $x_0 \in \text{Ann}(2\epsilon, 1)$ such that

$$(B_{4\epsilon}(x_0) \cap \Sigma_\infty) \cap (\cup_{j=1}^l \gamma_j) = \emptyset. \quad (4.28)$$

One can see Fig. 3 for intuition, where the case $l = 3$ was illustrated. Then (4.27) follows from the combination of (4.21) and (4.28). $\square$

Fix an interval $I = (-T, T)$ and choose $t \in I$. By (4.27), it is clear that the convergence from $\Sigma_{i,t}$ to Σ_∞ around point x_0 is smooth with multiplicity N_0 . Let α_{x_0} be the vertical line passing through $x_0 \in P$. For large i , we have

$$\Sigma_{i,t} \cap \alpha_{x_0} = \left\{ z_{i,t}^{(1)}, z_{i,t}^{(2)}, \dots, z_{i,t}^{(N_0)} \right\}, \quad (4.29)$$

where the points $z_{i,t}^{(j)}$ are ordered increasingly by their x_3 -coordinates. Namely, we have

$$x_3 \left(z_{i,t}^{(1)} \right) < x_3 \left(z_{i,t}^{(2)} \right) < \dots < x_3 \left(z_{i,t}^{(N_0)} \right). \quad (4.30)$$

Note that near x_0 , the smooth convergence guarantees that the normal directions of $\Sigma_{i,t}$ are almost perpendicular to the horizontal plane P . Therefore, the vertical line α_{x_0} intersects $\Sigma_{i,t}$ transversally. Together with the embeddedness of $\Sigma_{i,t}$, this forces that all the inequalities in (4.30) are strict. By Lemma 3.14, the multiplicity is independent of time. Therefore, for large i , (4.30) holds for

each $t \in (-T, T)$. In particular, we have

$$x_3 \left(z_{i,t}^{(j)} \right) < x_3 \left(z_{i,t}^{(k)} \right), \quad \forall t \in (-T, T); \quad \Leftrightarrow \quad x_3 \left(z_{i,0}^{(j)} \right) < x_3 \left(z_{i,0}^{(k)} \right). \quad (4.31)$$

We define

$$\Omega_\epsilon(t) := (\Sigma_\infty \cap B_{\epsilon^{-1}}(0)) \setminus \bigcup_{p \in \mathcal{S}_t} B_\epsilon(p), \quad (4.32)$$

$$\hat{\Omega}_\epsilon(t) := \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid (x_1, x_2, 0) \in \Omega_\epsilon(t), x_3 \in \mathbb{R}\}, \quad (4.33)$$

$$\Omega_\epsilon(I) := \bigcap_{t \in I} \Omega_\epsilon(t). \quad (4.34)$$

By its definition in (4.32), it is clear that the convergence around $\Omega_\epsilon(t)$ is smooth. Therefore, $\Sigma_{i,t} \cap \hat{\Omega}_\epsilon(t)$ is a multi-sheet graph of exactly N_0 sheet (c.f. Lemma 3.14). We define the connected component of $\Sigma_{i,t} \cap \hat{\Omega}_\epsilon(t)$ to be the j -th sheet if this component contains the point $z_{i,t}^{(j)}$. By (4.30), we know that this is well defined and the choice of j is independent of time. Furthermore, each component of $\Sigma_{i,t} \cap \hat{\Omega}_\epsilon(t)$ is a graph of smooth functions $u_i^{(j)}(\cdot, t)$ over $\Omega_\epsilon(t)$ such that

$$u_i^{(1)}(q, t) < u_i^{(2)}(q, t) < \cdots < u_i^{(N_0)}(q, t), \quad \forall q \in \Omega_\epsilon(t), \quad t \in (-T, T). \quad (4.35)$$

According to their construction, it is clear that

$$u_i^{(j)}(x'_0, t) = x_3 \left(z_{i,t}^{(j)} \right), \quad \forall j \in \{1, 2, \dots, N_0\},$$

where x'_0 is the orthogonal projection image of x_0 from $\mathbb{R}^3$ to the horizontal plane $P = \Sigma_\infty$. The j -th component of $\Sigma_{i,t} \cap \hat{\Omega}_\epsilon(t)$ is called top sheet if $j = N_0$ and it is called bottom sheet if $j = 1$. For simplicity of notation, let $u_i^+(x, t)$ and $u_i^-(x, t)$ be the functions representing the top and bottom sheets, i.e.,

$$u_i^+(x, t) := u_i^{(N_0)}(x, t), \quad u_i^-(x, t) := u_i^{(1)}(x, t). \quad (4.36)$$

Then we denote the graphs of $u_i^\pm(x, t)$ over Ω_ϵ by $\Sigma_{i,t}^\pm$ respectively. It is not hard to see that $u_i^\pm(x, t)$ satisfy an evolution equation determined by the rescaled mean curvature flow. We write down the details for the convenience of the readers.

Lemma 4.12 *The functions $u_i^+(x, t)$ and $u_i^-(x, t)$ satisfy the equation on $\Omega_\epsilon(I) \times I$*

$$\frac{\partial u}{\partial t} = \Delta_0 u - \frac{\nabla^2 u(\nabla u, \nabla u)}{1 + |\nabla u|^2} - \frac{1}{2} \langle x, \nabla u \rangle + \frac{u}{2}. \quad (4.37)$$

Proof Let $\Omega \subset P = \{(x_1, x_2, x_3) \mid x_3 = 0\}$. Consider the graph of the function $u : \Omega \times I \rightarrow \mathbb{R}$

$$\text{Graph}_u = \{(x_1, x_2, u(x_1, x_2, t)) \mid (x_1, x_2) \in \Omega, t \in I\}.$$

The upward pointing unit normal and the mean curvature are given by

$$\mathbf{n} = \frac{(-u_{x_1}, -u_{x_2}, 1)}{\sqrt{1 + |\nabla u|^2}}, \quad H = -\text{div}\left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}}\right) \quad (4.38)$$

respectively. Thus, we have

$$H - \frac{1}{2} \langle \mathbf{x}, \mathbf{n} \rangle = -\text{div}\left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}}\right) + \frac{x_1 u_{x_1} + x_2 u_{x_2} - u}{2\sqrt{1 + |\nabla u|^2}}, \quad (4.39)$$

where $\mathbf{x} = (x_1, x_2, u(x_1, x_2, t))$ denotes a point on Graph_u . On the other hand, we have

$$\left\langle \frac{\partial \mathbf{x}}{\partial t}, \mathbf{n} \right\rangle = \frac{\partial u}{\partial t} \frac{1}{\sqrt{1 + |\nabla u|^2}}. \quad (4.40)$$

Combining (4.39) and (4.40) with the equation of rescaled mean curvature flow (1.2), we have

$$\frac{\partial u}{\partial t} = \sqrt{1 + |\nabla u|^2} \text{div}\left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}}\right) - \frac{1}{2} \langle x, \nabla u \rangle + \frac{u}{2}, \quad (4.41)$$

where $x = (x_1, x_2, 0) \in \Omega$ and $\langle x, \nabla u \rangle = x_1 u_{x_1} + x_2 u_{x_2}$. Direct calculation shows that (4.41) implies (4.37). Note that $u_i^+(x, t)$ and $u_i^-(x, t)$ denote the graph functions of $\Sigma_{i,t}^+$ and $\Sigma_{i,t}^-$ respectively. Therefore, $u_i^+(x, t)$ and $u_i^-(x, t)$ satisfy (4.37). The lemma is proved. $\square$

Let $u_i = u_i^+ - u_i^-$ and $f_{i,s} = u_i^- + s u_i$ for $s \in [0, 1]$. Then $f_{i,0} = u_i^-$ and $f_{i,1} = u_i^+$. We define

$$G(s) := \frac{\nabla^2 f_{i,s}(\nabla f_{i,s}, \nabla f_{i,s})}{1 + |\nabla f_{i,s}|^2}.$$

Therefore, we have

$$G(1) - G(0) = \int_0^1 \frac{\partial G(s)}{\partial s} ds = a^{pq} u_{i,pq} + b^p u_{i,p}, \quad (4.42)$$

where

$$a^{pq} = \int_0^1 \left((1 + |\nabla f_{i,s}|)^{-1} \nabla_p f_{i,s} \nabla_q f_{i,s} \right) ds, \quad (4.43)$$

$$b^p = \int_0^1 \left(2(1 + |\nabla f_{i,s}|)^{-1} \nabla_k \nabla_p f_{i,s} \nabla_k f_{i,s} - 2(1 + |\nabla f_{i,s}|)^{-2} \nabla_p f_{i,s} \nabla^2 f_{i,s} (\nabla f_{i,s}, \nabla f_{i,s}) \right) ds. \quad (4.44)$$

Combining (4.37) with (4.42), we get the equation of u_i

$$\frac{\partial u_i}{\partial t} = \Delta_0 u_i - \frac{1}{2} \langle x, \nabla u_i \rangle + \frac{u_i}{2} - a^{pq} u_{i,pq} - b^p u_{i,p} \quad (4.45)$$

for any $(x, t) \in \Omega_\epsilon(I) \times I$. Since $\Sigma_{i,t}^+$ and $\Sigma_{i,t}^-$ converge locally smoothly to Σ_∞ as $i \rightarrow +\infty$, all derivatives of u_i^+ and u_i^- converge to zero on $\Omega_\epsilon(I) \times I$. Therefore, by the expression (4.43)–(4.44) all $C^{k,\alpha}$ norms of a^{pq} and b^p are uniformly bounded when t_i is large and tends to zero as $t_i \rightarrow +\infty$.

Clearly, u_i are positive solutions satisfying $u_i \rightarrow 0$. We shall normalize u_i to obtain w_i which satisfies the linearization of the equation satisfied by u_i . For this purpose, we need a space–time base point with uniform regularity. Note that the space–time singular set $\mathcal{S}$ has good regularity (4.21) and the number of points in $\mathcal{S}_0$ is uniformly bounded by (4.22). It is not hard to see that

$$\Omega_{\frac{\epsilon}{8}}(0) \subset \Omega_{\frac{\epsilon}{10}}(t), \quad \forall t \in [-2\epsilon', 2\epsilon'] \quad (4.46)$$

for some $\epsilon' = \epsilon'(S_0, \epsilon) > 0$. Then (x_0, ϵ') is a good space–time base point. We now normalize the function u_i by

$$w_i(x, t) := \frac{u_i(x, t)}{u_i(x_0, \epsilon')}, \quad \forall (x, t) \in \Omega_\epsilon(I) \times I. \quad (4.47)$$

Then $w_i(x, t)$ is a positive function on $\Omega_\epsilon(I) \times I$ satisfying $w_i(x_0, \epsilon') = 1$ and

$$\frac{\partial w_i}{\partial t} = \Delta_0 w_i - \frac{1}{2} \langle x, \nabla w_i \rangle + \frac{w_i}{2} - a^{pq} w_{i,pq} - b^p w_{i,p}, \quad (4.48)$$

where a^{pq} and b^p are defined by (4.43) and (4.44).

The estimate of $|\mathbf{TN}|$ in (4.19) provides information on the growth rate of w_i , through the application of the following Lemma. It basically means that the volume of the thin part is almost the volume of the region between the top and bottom sheets.

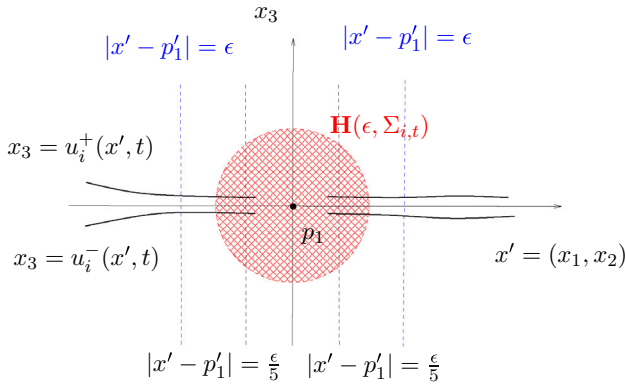


Fig. 4 Thin part is almost the region between the top and bottom sheets

Lemma 4.13 For large t_i , we have

$$\int_{\Omega_\epsilon(t)} u_i(x, t) d\mu_\infty \leq |\mathbf{TN}(\epsilon, \Sigma_{i,t})| \leq \int_{\Omega_\epsilon(t)} u_i(x, t) d\mu_\infty, \quad (4.49)$$

where $d\mu_\infty$ denotes the standard volume form of Σ_∞ .

Proof For each t and large t_i , we define

$$\Phi(\epsilon, \Sigma_{i,t}) = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x' := (x_1, x_2, 0) \in \Omega_\epsilon(t), u_i^-(x', t) \leq x_3 \leq u_i^+(x', t)\}.$$

Then the Euclidean volume of $\Phi(\epsilon, \Sigma_{i,t})$ can be calculated by

$$|\Phi(\epsilon, \Sigma_{i,t})| = \int_{\Omega_\epsilon(t)} u_i(x, t) d\mu_\infty. \quad (4.50)$$

We claim that

$$\Phi(\epsilon, \Sigma_{i,t}) \subset \mathbf{TN}(\epsilon, \Sigma_{i,t}) \subset \Phi\left(\frac{\epsilon}{5}, \Sigma_{i,t}\right). \quad (4.51)$$

One can check Fig. 4 for intuition, where the case $\mathcal{S}_t = \{p_1\}$ is addressed and p'_1 is the (x_1, x_2) coordinate of the point p_1 .

Now we prove (4.51) rigorously. Note that $\Sigma_{i,t}$ converges locally smoothly to Σ_∞ away from $\mathcal{S}_t$. For each t , the set $\mathbf{S}(\epsilon, \Sigma_{i,t})$ converges to $\mathcal{S}_t$. So for large i , we have

$$\mathbf{S}(\epsilon, \Sigma_{i,t}) \subset B_{\frac{\epsilon}{10}}(\mathcal{S}_t) = \cup_{p \in \mathcal{S}_t} B_{\frac{\epsilon}{10}}(p).$$

By the definition of $\mathbf{H}$ in Definition 4.4, we have

$$\mathbf{H}(\epsilon, \Sigma_{i,t}) \subset B_{\frac{3}{4}\epsilon}(\mathcal{S}_t) \cap B_{\epsilon^{-1}}(0). \quad (4.52)$$

For any $x = (x_1, x_2, x_3) \in \Phi(\epsilon, \Sigma_{i,t})$, we have $(x_1, x_2, 0) \in \Omega_\epsilon(t)$, which implies that $(x_1, x_2, x_3) \notin B_\epsilon(\mathcal{S}_t)$. Therefore, $x \notin \mathbf{H}(\epsilon, \Sigma_{i,t})$ and we have $\Phi(\epsilon, \Sigma_{i,t}) \subset \mathbf{TN}(\epsilon, \Sigma_{i,t})$ when t_i is large. On the other hand, for any $x \in \mathbf{TN}(\epsilon, \Sigma_{i,t})$, we have $x \notin \mathbf{TK}(\epsilon, \Sigma_{i,t})$ and $x \notin \mathbf{H}(\epsilon, \Sigma_{i,t})$, which implies that $d(x, \mathbf{S}) \geq \frac{\epsilon}{2}$ by the definition of $\mathbf{H}$. Since $\mathcal{S}_t$ consists of finite many points, $d(x, \mathcal{S}_t)$ is attained by some point $q_0 \in \mathcal{S}_t$. Thus, we have

$$d(x, \mathcal{S}_t) = \min_{y \in \mathcal{S}_t} d(x, y) = d(x, q_0) \geq \max_{z \in \mathbf{S} \cap B_{\frac{\epsilon}{10}}(q_0)} (d(x, z) - d(z, q_0)) \geq \frac{\epsilon}{4}, \quad (4.53)$$

where we used the fact that $d(x, z) \geq \frac{\epsilon}{2}$ and $d(z, q_0) \leq \frac{\epsilon}{10}$. Moreover, since $x \notin \mathbf{TK}(\epsilon, \Sigma_{i,t})$, x is sufficiently close to Σ_∞ . This together with (4.53) implies that $x \in \Phi(\frac{\epsilon}{5}, \Sigma_{i,t})$ when i is large. Therefore, (4.51) is proved. Thus, the inequality (4.49) follows from (4.50) and (4.51). The lemma is proved. $\square$

Combining the previous discussion, we obtain some uniform estimates on the functions w_i .

Lemma 4.14 *For any compact set $K \Subset \text{Ann}(2\epsilon, \frac{1}{2}\epsilon^{-1})$, $T > t_\epsilon + 2$, there exist two constants*

$$C_1 = C_1(\epsilon, K, \mathcal{S}_0, x_0) > 0, \quad C_2 = C_2(\epsilon, \mathcal{S}_0, T, x_0) > 0$$

such that for large t_i , we have

$$C_2(\epsilon, \mathcal{S}_0, T, x_0) < w_i(x, t) \leq C_1(\epsilon, K, \mathcal{S}_0, x_0), \quad \forall (x, t) \in K \times [t_\epsilon + 1, T - 1]. \quad (4.54)$$

Moreover, at time $t = t_\epsilon + 1$ there exists $C_3 = C_3(\epsilon, \mathcal{S}_0, x_0) > 0$ independent of T such that

$$w_i(x, t_\epsilon + 1) \geq C_3(\epsilon, \mathcal{S}_0, x_0) > 0, \quad \forall x \in K. \quad (4.55)$$

Proof We divide the proof into several steps.

Step 1. $w_i(x, 0)$ is bounded from above on K . By the definition of ϵ' in (4.46), all points of $\Omega_{\frac{\epsilon}{8}}(0)$ are regular for any $t \in [-2\epsilon', 2\epsilon']$. Thus, the Eq. (4.48) of w_i can be written as

$$\frac{\partial w_i}{\partial t} = \tilde{a}^{pq} w_{i,pq} + \tilde{b}^p w_{i,p} + \frac{1}{2} w_i, \quad \forall (x, t) \in \Omega_{\frac{\epsilon}{8}}(0) \times (-2\epsilon', 2\epsilon'), \quad (4.56)$$

where $\tilde{a}^{pq} = \delta^{pq} - a^{pq}$ and $\tilde{b}^p = -\frac{1}{2}x^p - b^p$. Here $\delta^{pq} = 0$ if $p \neq q$ and $\delta^{pq} = 1$ if $p = q$. Since the sequence $\{\Sigma_{i,t}, -T < t < T\}$ converges locally smoothly to the plane P away from the singular set $\mathcal{S}$, the coefficients a^{pq} and b^p tend to zero on $\Omega_{\frac{\epsilon}{8}}(0) \times (-2\epsilon', 2\epsilon')$. Thus, $\tilde{a}^{pq}$ and $\tilde{b}^p$ satisfy the conditions (A.2)–(A.3) in “Appendix A”. We set

$$\Omega := \Omega_{\frac{\epsilon}{8}}(0), \quad \Omega' := \Omega_{\frac{\epsilon}{5}}(0), \quad \Omega'' := \Omega_{\frac{\epsilon}{6}}(0). \quad (4.57)$$

It follows from the definition (4.32) that we have the relationship

$$\Omega' \subseteq \Omega'' \subseteq \Omega. \quad (4.58)$$

Since $\overline{\Omega'}$ is compact, we can cover Ω' by finite many balls contained in Ω'' with radius $r = \frac{\epsilon}{100}$ and the number of these balls is bounded by a constant depending only on ϵ . Moreover, Ω'' has a positive distance $\delta = \delta(\epsilon) > 0$ from the boundary of Ω . Thus, applying Theorem A.4 for such $\Omega, \Omega', \Omega''$ and the time interval $(-2\epsilon', 2\epsilon')$, we have

$$w_i(x, 0) \leq C(\epsilon, \epsilon', x_0) w_i(x_0, \epsilon') = C(\epsilon, x_0, S_0), \quad \forall x \in \Omega'. \quad (4.59)$$

Here $C(\epsilon, x_0, S_0)$ means that C depends only on ϵ, x_0, S_0 .

Step 2. $w_i(x, t)$ is bounded from above on $K \times [t_\epsilon + 1, T - 1]$. Integrating both sides of (4.59) on Ω' , we have

$$\int_{\Omega_{\frac{\epsilon}{5}}(0)} w_i(x, 0) \leq C(\epsilon, x_0, S_0). \quad (4.60)$$

By Lemma 4.8 and Lemma 4.13, for large t_i and any $t \in [0, T) \subset [0, i]$ we have

$$\begin{aligned} \int_{\Omega_{\epsilon}(t)} u_i(x, t) d\mu_{\infty} &\leq |\mathbf{TN}(\epsilon, \Sigma_{t_i+t})| \leq 2|\mathbf{TN}(\epsilon, \Sigma_{t_i})| \\ &\leq 2 \int_{\Omega_{\frac{\epsilon}{5}}(0)} u_i(x, 0) d\mu_{\infty}. \end{aligned} \quad (4.61)$$

Combining (4.61) with the definition (4.47) of w_i , for large t_i and any $t \in [0, T)$ we have the following inequality

$$\int_{\Omega_{\epsilon}(t)} w_i(x, t) d\mu_{\infty} \leq 2 \int_{\Omega_{\frac{\epsilon}{5}}(0)} w_i(x, 0) d\mu_{\infty} \leq C(\epsilon, x_0, S_0), \quad (4.62)$$

where we used (4.60) in the last inequality. Note that by Lemma 4.10, for $t \in [t_\epsilon, T)$ we have $\Omega_\epsilon(t) = B_{\epsilon^{-1}}(0) \cap \Sigma_\infty$ if $0 \notin S_0$ and $\Omega_\epsilon(t) = \text{Ann}(\epsilon, \epsilon^{-1})$ if $0 \in S_0$. Let

$$\Omega := \text{Ann}(\epsilon, \epsilon^{-1}), \quad \Omega' := \text{Ann}(2\epsilon, \frac{1}{2}\epsilon^{-1}), \quad \Omega'' := \text{Ann}(\frac{3}{2}\epsilon, \frac{2}{3}\epsilon^{-1}). \quad (4.63)$$

Then we have $\Omega' \subseteq \Omega'' \subseteq \Omega$. Moreover, Ω' can be covered by finite many balls contained in Ω'' with radius $r = \frac{\epsilon}{100}$ and the number of these balls is bounded by a constant depending only on ϵ . For any $K \subset \subset \Omega'$, applying Theorem A.4 for such $\Omega, \Omega', \Omega''$ and the interval $[t_\epsilon, T - 1]$ as in Step 1, we have

$$w_i(x, t) \leq C(\epsilon)w_i(y, t + \frac{1}{2}), \quad \forall x, y \in K, \quad t \in [t_\epsilon + 1, T - 1]. \quad (4.64)$$

Integrating both sides of (4.64) for $y \in K$, we have

$$\begin{aligned} w_i(x, t) &\leq C(\epsilon, K) \int_K w_i(y, t + \frac{1}{2}) d\mu_\infty(y) \\ &\leq C(\epsilon, K) \int_{\Omega_\epsilon(t + \frac{1}{2})} w_i(y, t + \frac{1}{2}) d\mu_\infty(y) \\ &\leq C(\epsilon, x_0, S_0, K), \quad \forall (x, t) \in K \times [t_\epsilon + 1, T - 1], \end{aligned} \quad (4.65)$$

where we used (4.62) and the fact that

$$K \subset \subset \text{Ann}(2\epsilon, \frac{1}{2}\epsilon^{-1}) \subset \text{Ann}(\epsilon, \epsilon^{-1}) \subset \Omega_\epsilon(t), \quad \forall t \in [t_\epsilon, T).$$

Therefore, $w_i(x, t)$ is bounded from above.

Step 3. $w_i(x_0, t_\epsilon + \frac{1}{2})$ is bounded from below by a constant independent of i . By (4.27) and (4.34), we have $(B_\epsilon(x_0) \cap \Sigma_\infty) \subset \Omega_\epsilon((-T, T))$. Therefore, applying Theorem A.3 for $(B_\epsilon(x_0) \cap \Sigma_\infty) \times [0, t_\epsilon + 1]$ we have

$$w_i(x_0, t_\epsilon + \frac{1}{2}) \geq C(\epsilon, x_0, S_0)w_i(x_0, \epsilon') = C(\epsilon, x_0, S_0), \quad (4.66)$$

where we used $w_i(x_0, \epsilon') = 1$. Thus, $w_i(x_0, t_\epsilon + \frac{1}{2})$ is bounded from below.

Step 4. $w_i(x, t)$ has a positive lower bound on $K \times [t_\epsilon + 1, T - 1]$. Note that by Lemma 4.11 we have $x_0 \in \text{Ann}(2\epsilon, \frac{1}{2}\epsilon^{-1})$. We set Ω, Ω' and Ω'' as in (4.63) and we have $K \cup \{x_0\} \subset \Omega'$. Thus, we apply Theorem A.4 for such

Ω, Ω' and the interval $[t_\epsilon, T - 1]$ to get

$$\begin{aligned} w_i(x, t) &\geq C(\epsilon, x_0, t - (t_\epsilon + \frac{1}{2}), \mathcal{S}_0) w_i(x_0, t_\epsilon + \frac{1}{2}), \\ \forall (x, t) &\in K \times [t_\epsilon + 1, T - 1]. \end{aligned} \quad (4.67)$$

Combining (4.67) with (4.66), we have

$$w_i(x, t) \geq C(\epsilon, x_0, T, \mathcal{S}_0), \quad \forall (x, t) \in K \times [t_\epsilon + 1, T - 1]. \quad (4.68)$$

In particular, if we take $t = t_\epsilon + 1$ in (4.67), then we have

$$w_i(x, t_\epsilon + 1) \geq C(\epsilon, x_0, \mathcal{S}_0) w_i(x_0, t_\epsilon + \frac{1}{2}) \geq C(\epsilon, x_0, \mathcal{S}_0), \quad \forall x \in K. \quad (4.69)$$

Note that the constant C in (4.69) doesn't depend on T . The lemma is proved. $\square$

Lemma 4.15 *The same conditions as in Lemma 4.14. As $t_i \rightarrow +\infty$, we can take a subsequence of the functions $w_i(x, t)$ such that it converges in C^2 topology on $K \times [t_\epsilon + 1, T - 1]$, where K is any compact set satisfying $K \subset \subset \text{Ann}(2\epsilon, \frac{1}{2}\epsilon^{-1})$, to a positive function $w(x, t)$ satisfying*

$$\frac{\partial w}{\partial t} = Lw := \Delta_0 w - \frac{1}{2} \langle x, \nabla w \rangle + \frac{1}{2} w, \quad \forall (x, t) \in K \times [t_\epsilon + 1, T - 1]. \quad (4.70)$$

Moreover, there exist constants $C_1(\epsilon, K, \mathcal{S}_0, x_0), C_2(\epsilon, \mathcal{S}_0, T, x_0) > 0$ such that

$$C_2(\epsilon, \mathcal{S}_0, T, x_0) \leq w(x, t) \leq C_1(\epsilon, K, \mathcal{S}_0, x_0), \quad \forall (x, t) \in K \times [t_\epsilon + 1, T - 1], \quad (4.71)$$

and at time $t = t_\epsilon + 1$ there exists $C_3 = C_3(\epsilon, \mathcal{S}_0, x_0) > 0$ independent of T such that

$$w(x, t_\epsilon + 1) \geq C_3(\epsilon, \mathcal{S}_0, x_0) > 0, \quad \forall x \in K. \quad (4.72)$$

Proof Let $I = [t_\epsilon + 1, T - 1]$. By Lemma 4.14, the function $w_i(x, t)$ is bounded from above and below by some constants independent of i . Since the function w_i satisfies the Eq. (4.56) on any $K \times I$, we can use the interior Hölder estimate (c.f. Theorem 4.9 of [37], or Theorem B.1 in “Appendix B”) to get the local space–time $C^{2,\alpha}$ estimates, which implies that a subsequence converges uniformly in C^2 to a positive function $w(x, t)$ on $(x, t) \in K \times I$ satisfying (4.70)–(4.72). The lemma is proved. $\square$

Now we are able to finish the proof of Proposition 4.9.

Proof of Proposition 4.9 Note that by Property (4) in the beginning of this subsection, P and S_0 are independent of T , and by Lemma 4.10 we see that t_ϵ is also independent of T . Moreover, (4.71) implies that the space-times $C^{2,\alpha}$ norm of w is bounded by a constant independent of T on $K \times [t_\epsilon + 1, T - 1]$ (c.f. Theorem 4.9 of [37], or Theorem B.1 in “Appendix B”). Thus, we can take $T \rightarrow \infty$ in Lemma 4.15 and obtain a function, still denoted by w , defined on $\text{Ann}(3\epsilon, \frac{1}{3}\epsilon^{-1}) \times [t_\epsilon + 1, \infty)$ satisfying (4.70), (4.71) and (4.72). Replacing ϵ by $\frac{1}{3}\epsilon$ and t by $t - t_\epsilon - 1$, we know that the function w is defined on $\text{Ann}(\epsilon, \epsilon^{-1}) \times [0, \infty)$ and satisfies (4.16), (4.17) and (4.18). The proposition is proved. $\square$

4.4 Proof of the multiplicity-one convergence

In this subsection, we shall show Theorem 4.1, i.e., the limit plane is multiplicity one, under the special condition H decays exponentially fast. Our argument largely follows that of Colding-Minicozzi [15]. However, some new technical difficulties need to be addressed. For example, the proof of the limit surface to be L -stable (c.f. Lemma 4.16) is technically different from the classical one. Due to the loss of general self-shinker equation, we need to delicately choose subsequence and apply parabolic version of maximum principle. The final contradiction relies heavily on the existence of the auxiliary function w constructed in Proposition 4.9.

Lemma 4.16 *Let P be the horizontal plane defined in (4.14). For any function $\phi \in W_c^{1,2}(P \setminus \{0\})$, we have*

$$-\int_P (\phi L\phi) e^{-\frac{|x|^2}{4}} \geq 0. \quad (4.73)$$

Here $W_c^{1,2}(P \setminus \{0\})$ denotes the set of all functions $\phi \in W^{1,2}(P)$ with compact support in $P \setminus \{0\}$, and L is the operator defined by (4.70) on the plane P .

Proof Given a function $\phi \in W_c^{1,2}(P \setminus \{0\})$, there exists $\epsilon > 0$ such that

$$\overline{\text{Supp}(\phi)} \subset \text{Ann}(2\epsilon, \frac{1}{2}\epsilon^{-1}).$$

For such ϵ , we choose $\{t_i\}$ as in Lemma 4.8 and we denote by Σ_∞ the limit plane of the sequence $\{\Sigma_{t_i}\}$. Up to rotation in $\mathbb{R}^3$, we may assume Σ_∞ to be P . In light of Proposition 4.9, we obtain a positive function w defined on $\text{Ann}(\epsilon, \epsilon^{-1}) \times [0, \infty)$ satisfying (4.16), (4.17) and (4.18). Let $v := \log w$. It follows from (4.16) that v satisfies the equation

$$\frac{\partial v}{\partial t} = \Delta_0 v + \frac{1}{2} - \frac{1}{2} \langle x, \nabla v \rangle + |\nabla v|^2, \quad \forall (x, t) \in \text{Ann}(\epsilon, \epsilon^{-1}) \times [0, \infty).$$

Then integration by parts implies that

$$\begin{aligned} 0 &= \int_P \operatorname{div} \left(\phi^2 e^{-\frac{|x|^2}{4}} \nabla v \right) \\ &= \int_P \left(2\phi \langle \nabla \phi, \nabla v \rangle + \left(\frac{\partial v}{\partial t} - \frac{1}{2} - |\nabla v|^2 \right) \phi^2 \right) e^{-\frac{|x|^2}{4}} \\ &\leq \int_P \left(|\nabla \phi|^2 - \frac{1}{2} \phi^2 + \frac{\partial v}{\partial t} \phi^2 \right) e^{-\frac{|x|^2}{4}}. \end{aligned}$$

The above inequality can be rewritten as

$$\begin{aligned} - \int_P (\phi L\phi) e^{-\frac{|x|^2}{4}} &= \int_P \left(|\nabla \phi|^2 - \frac{1}{2} \phi^2 \right) e^{-\frac{|x|^2}{4}} \\ &\geq - \int_P \frac{\partial v}{\partial t} \phi^2 e^{-\frac{|x|^2}{4}} = - \frac{d}{dt} \int_P v \phi^2 e^{-\frac{|x|^2}{4}}. \end{aligned} \quad (4.74)$$

Fix $T > 0$. Integrating both sides of (4.74), we have

$$\begin{aligned} - \int_0^T dt \int_P (\phi L\phi) e^{-\frac{|x|^2}{4}} &\geq \int_P v \phi^2 e^{-\frac{|x|^2}{4}} \Big|_{t=0} - \int_P v \phi^2 e^{-\frac{|x|^2}{4}} \Big|_{t=T} \\ &\geq -C, \end{aligned} \quad (4.75)$$

where C is independent of T by (4.17) and (4.18) in Proposition 4.9. Thus, we have

$$- \int_P (\phi L\phi) e^{-\frac{|x|^2}{4}} \geq -\frac{C}{T}. \quad (4.76)$$

Letting $T \rightarrow +\infty$ in (4.76) we obtain (4.73). The proof of Lemma 4.16 is complete. $\square$

Following the argument of Gulliver-Lawson [21], we show that the plane is L -stable across the singular set $\mathcal{S}$. In other words, we have the following Lemma.

Lemma 4.17 *The same assumption as in Lemma 4.16. For any smooth function ϕ with compact support in the plane P , we have*

$$- \int_P (\phi L\phi) e^{-\frac{|x|^2}{4}} \geq 0. \quad (4.77)$$

Proof Choose $0 < \delta < R < 1$ and define the function

$$\eta(x) = \begin{cases} \frac{\log R}{\log |x|}, & 0 < |x| < R, \\ 1, & |x| \geq R \end{cases}$$

and the function $\beta(x) = \beta(|x|)$ such that $\beta(x) = 0$ for $|x| < \frac{\delta}{2}$, $\beta(x) = 1$ for $|x| \geq \delta$, $0 \leq \beta(x) \leq 1$ and $|\nabla \beta| \leq \frac{3}{\delta}$. Suppose that ϕ is any function with compact support on P . Moreover, we can check that

$$\int_P |\nabla \eta|^2 e^{-\frac{|x|^2}{4}} \leq \frac{C}{|\log R|}, \quad \text{and} \quad \int_P |\nabla \beta|^2 e^{-\frac{|x|^2}{4}} \leq C, \quad (4.78)$$

where C are universal constants. Note that $\eta\beta\phi$ has compact support in $P \setminus \{0\}$. Using the inequality

$$(a + b + c)^2 \leq 2\left(1 + \frac{1}{\tau}\right)(a^2 + b^2) + (1 + \tau)c^2, \quad \forall \tau > 0,$$

we have

$$|\nabla(\eta\beta\phi)|^2 \leq 2\left(1 + \frac{1}{\tau}\right)\phi^2\left(\beta^2|\nabla\eta|^2 + \eta^2|\nabla\beta|^2\right) + (1 + \tau)\eta^2\beta^2|\nabla\phi|^2. \quad (4.79)$$

We normalize ϕ such that $|\phi| \leq 1$. Note that $\eta\beta\phi \in W_c^{1,2}(P \setminus \{0\})$. Combining (4.79) with Lemma 4.16, we have

$$\begin{aligned} 0 &\leq - \int_P \left(\eta\beta\phi \cdot L(\eta\beta\phi) \right) e^{-\frac{|x|^2}{4}} \\ &= \int_P \left(|\nabla(\eta\beta\phi)|^2 - \frac{1}{2}(\eta\beta\phi)^2 \right) e^{-\frac{|x|^2}{4}} \\ &\leq (1 + \tau) \int_P \left(|\nabla\phi|^2 - \frac{1}{2}\phi^2 \right) e^{-\frac{|x|^2}{4}} + C\left(1 + \frac{1}{\tau}\right) \frac{1}{|\log R|} + C\left(\frac{\log R}{\log \delta}\right)^2 \\ &\quad + \frac{1}{2} \int_P \left(1 + \tau - (\eta\beta)^2 \right) \phi^2 e^{-\frac{|x|^2}{4}}, \end{aligned}$$

where C are universal constants. Taking $\delta \rightarrow 0$ and then $R \rightarrow 0$, we have

$$0 \leq (1 + \tau) \int_P \left(|\nabla\phi|^2 - \frac{1}{2}\phi^2 \right) e^{-\frac{|x|^2}{4}} + \frac{\tau}{2} \int_P \phi^2 e^{-\frac{|x|^2}{4}}.$$

Letting $\tau \rightarrow 0$, we get the inequality

$$\int_P \left(|\nabla\phi|^2 - \frac{1}{2}\phi^2 \right) e^{-\frac{|x|^2}{4}} \geq 0. \quad (4.80)$$

The lemma is proved. $\square$

The rest of the argument is the same as that in Colding-Minicozzi [15]. Let $\rho : \mathbb{R} \rightarrow [0, 1]$ be a smooth cutoff function with $\rho(s) = 1$ for $s \leq R$, $\rho = 0$ for $s \geq R + 1$, $0 \leq \rho(s) \leq 1$ and $|\rho'(s)| \leq 2$ for $s \in \mathbb{R}$. Let $\phi(x) = \rho(|x|)$ for any $x \in P$, where P is defined in (4.14). Then Lemma 4.17 implies that

$$\begin{aligned} 0 &\leq - \int_P (\phi L \phi) e^{-\frac{|x|^2}{4}} = \int_P \left(|\nabla \phi|^2 - \frac{1}{2} \phi^2 \right) e^{-\frac{|x|^2}{4}} \\ &\leq \int_P \left((\rho'(|x|))^2 - \frac{1}{2} \rho(|x|)^2 \right) e^{-\frac{|x|^2}{4}} \\ &\leq 4 \int_{P \setminus B_R(0)} e^{-\frac{|x|^2}{4}} - \frac{1}{2} \int_{B_R(0) \cap P} e^{-\frac{|x|^2}{4}}. \end{aligned} \quad (4.81)$$

Note that the right hand side of (4.81) is negative when R is sufficiently large. Therefore, we get a contradiction. Thus, for the sequence of times $\{t_i\}$ in Lemma 4.8, the multiplicity of the convergence of $\{(\Sigma, \mathbf{x}(t_i + t)), -T < t < T\}$ is one and the convergence is smooth everywhere. Theorem 4.1 is proved.

5 Proof of the extension theorem

For the convenience of readers, we copy down the statement of our main extension theorem, i.e., Theorem 1.1 as follows.

Theorem 5.1 *If $\mathbf{x}(p, t) : \Sigma^2 \rightarrow \mathbb{R}^3 (t \in [0, T))$ is a closed smooth embedded mean curvature flow with the first singular time $T < +\infty$, then*

$$\sup_{\Sigma \times [0, T)} |H|(p, t) = +\infty.$$

Proof Suppose not, we can find (x_0, T) such that the mean curvature flow $\mathbf{x}(p, t)$ blows up at $x_0 \in \mathbb{R}^3$ at time T with

$$\Lambda_0 := \sup_{\Sigma \times [0, T)} |H|(p, t) < +\infty. \quad (5.1)$$

Then Corollary 3.6 of [18] implies that for all $t < T$ we have

$$d(\Sigma_t, x_0) \leq 2\sqrt{T - t}, \quad (5.2)$$

where $d(\Sigma_t, x_0)$ denotes the Euclidean distance from the point x_0 to the surface Σ_t . We can rescale the flow Σ_t by

$$s = -\log(T - t), \quad \tilde{\Sigma}_s = e^{\frac{s}{2}} (\Sigma_{T-e^{-s}} - x_0)$$

such that the flow $\{(\tilde{\Sigma}_s, \tilde{\mathbf{x}}(p, s)), -\log T \leq s < +\infty\}$ satisfies the following properties:

(1) $\tilde{\mathbf{x}}(p, s)$ satisfies the equation

$$\left(\frac{\partial \tilde{\mathbf{x}}}{\partial s}\right)^\perp = -\left(\tilde{H} - \frac{1}{2}\langle \tilde{\mathbf{x}}, \mathbf{n} \rangle\right)\mathbf{n}; \quad (5.3)$$

(2) the mean curvature of $\tilde{\Sigma}_s$ satisfies $|\tilde{H}(p, s)| \leq \Lambda_0 e^{-\frac{s}{2}}$ for some $\Lambda_0 > 0$;
 (3) $d(\tilde{\Sigma}_s, 0) \leq 2$.

By Theorem 4.1, there exists a sequence of times $s_i \rightarrow +\infty$ such that $\tilde{\Sigma}_{s_i}$ converges smoothly to a plane passing through the origin with multiplicity one. Consider the heat-kernel-type function

$$\Phi_{(x_0, T)}(x, t) = \frac{1}{4\pi(T-t)} e^{-\frac{|x-x_0|^2}{4(T-t)}}, \quad \forall (x, t) \in \Sigma_t \times [0, T).$$

Huisken's monotonicity formula(c.f. Theorem 3.1 in [26]) implies that

$$\begin{aligned} \Theta(\Sigma_t, x_0, T) &:= \lim_{t \rightarrow T} \int_{\Sigma_t} \Phi_{(x_0, T)}(x, t) d\mu_t \\ &= \lim_{s_i \rightarrow +\infty} \frac{1}{4\pi} \int_{\tilde{\Sigma}_{s_i}} e^{-\frac{|x|^2}{4}} d\tilde{\mu}_{s_i} = 1, \end{aligned}$$

which implies that (x_0, T) is a regular point by Theorem 3.1 of B. White [47]. Thus, the unnormalized mean curvature flow $\{(\Sigma, \mathbf{x}(t)), 0 \leq t < T\}$ cannot blow up at (x_0, T) , which contradicts our assumption. The theorem is proved. $\square$

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Appendix A The parabolic Harnack inequality

In this ‘‘Appendix’’, we include the parabolic Harnack inequality from Krylov-Safonov [31, 32]. First, we introduce some notations. Let $x = (x^1, x^2, \dots, x^n) \in \mathbb{R}^n$. Denote

$$|x| = \left(\sum_{i=1}^n (x^i)^2\right)^{\frac{1}{2}}, \quad B_R(x) = \{y \in \mathbb{R}^n \mid |x - y| < R\},$$

$$Q(\theta, R) = B_R(0) \times (0, \theta R^2).$$

Consider the parabolic operator

$$Lu = -\frac{\partial u}{\partial t} + a^{ij}(x, t)u_{ij} + b^i(x, t)u_i - c(x, t)u, \quad (\text{A.1})$$

where the coefficients are measurable and satisfy the conditions

$$\mu|\xi|^2 \leq a^{ij}(x, t)\xi_i\xi_j \leq \frac{1}{\mu}|\xi|^2, \quad (\text{A.2})$$

$$|b(x, t)| \leq \frac{1}{\mu}, \quad (\text{A.3})$$

$$0 \leq c(x, t) \leq \frac{1}{\mu}. \quad (\text{A.4})$$

Here $b(x, t) = (b^1(x, t), \dots, b^n(x, t))$. Then we have

Theorem A.1 (Theorem 1.1 of [31, 32]) *Suppose the operator L in (A.1) satisfies the conditions (A.2)–(A.4). Let $\theta > 1$, $R \leq 2$, $u \in W_{n+1}^{1,2}(Q(\theta, R))$, $u \geq 0$ in $Q(\theta, R)$, and $Lu = 0$ on $Q(\theta, R)$. Then there exists a constant C , depending only on θ , μ and n , such that*

$$u(0, R^2) \leq C u(x, \theta R^2), \quad \forall x \in B_{\frac{R}{2}}(0). \quad (\text{A.5})$$

Moreover, when $\frac{1}{\theta-1}$ and $\frac{1}{\mu}$ vary within finite bounds, C also varies within finite bounds.

Note that in our case the Eq. (4.48) doesn't satisfy the assumption that $c(x, t) \geq 0$ in (A.4). Therefore, we cannot use Theorem A.1 directly. However, the Harnack inequality still works when $c(x, t)$ is a constant. Namely, we have

Theorem A.2 *Let $\theta > 1$, $R \leq 2$. Suppose that $u(x, t) \in W_{n+1}^{1,2}(Q(\theta, R))$ is a nonnegative solution to the equation*

$$Lu = -\frac{\partial u}{\partial t} + a^{ij}(x, t)u_{ij} + b^i(x, t)u_i + cu = 0, \quad (\text{A.6})$$

where c is a constant and the coefficients satisfy (A.2)–(A.3). Then there exists a constant C , depending only on θ , μ , c and n , such that

$$u(0, R^2) \leq C u(x, \theta R^2), \quad \forall |x| < \frac{1}{2}R. \quad (\text{A.7})$$

Proof Since $u(x, t)$ is a solution of (A.6) and c is a constant, the function $v(x, t) = e^{-ct}u$ satisfies

$$-\frac{\partial v}{\partial t} + a^{ij}(x, t)v_{ij} + b^i(x, t)v_i = 0. \quad (\text{A.8})$$

Applying Theorem A.1 to the Eq. (A.8), we have

$$v(0, R^2) \leq C v(x, \theta R^2), \quad \forall |x| < \frac{1}{2}R,$$

where C depends only on θ, μ and n . Thus, for any $x \in B_{\frac{R}{2}}(0)$ we have

$$u(0, R^2) \leq C e^{-c(\theta-1)R^2} u(x, \theta R^2) \leq C' u(x, \theta R^2),$$

where C' depends only on θ, μ, c and n . Here we used $R \leq 2$ by the assumption. The theorem is proved. $\square$

We generalize Theorem A.2 to a general bounded domain in $\mathbb{R}^n$.

Theorem A.3 *Let Ω be a bounded domain in $\mathbb{R}^n$. Suppose that $u(x, t) \in W_{n+1}^{1,2}(\Omega \times (0, T))$ is a nonnegative solution to the equation*

$$Lu = -\frac{\partial u}{\partial t} + a^{ij}(x, t)u_{ij} + b^i(x, t)u_i + cu = 0, \quad (\text{A.9})$$

where c is a constant and the coefficients satisfy (A.2)–(A.3) for a constant $\mu > 0$. For any s, t satisfying $0 < s < t < T$ and any $x, y \in \Omega$ with the following properties

- (1). x and y can be connected by a line segment γ with the length $|x - y| \leq l$;
- (2). Each point in γ has a positive distance at least $\delta > 0$ from the boundary of Ω ;
- (3) s and t satisfy $T_1 \leq t - s \leq T_2$ for some $T_1, T_2 > 0$;

we have

$$u(y, s) \leq C u(x, t), \quad (\text{A.10})$$

where C depends only on $c, n, \mu, \min\{s, \delta^2\}, l, T_1$ and T_2 .

Proof Let γ be the line segment with the property (1) and (2) connecting x and y . We set

$$p_0 = y, \quad p_N = x, \quad p_i = p_0 + \frac{x - y}{N}i \in \gamma$$

for any $0 \leq i \leq N$. Here we choose N to be the smallest integer satisfying

$$N > \max \left\{ \frac{2(t-s)}{s}, \frac{l}{\min\{\frac{\sqrt{s}}{4}, \frac{\delta}{4}\}} \right\}. \quad (\text{A.11})$$

We define

$$R = \frac{2l}{N}, \quad \theta = 1 + \frac{t-s}{R^2 N}. \quad (\text{A.12})$$

We can check that $R \leq \frac{\delta}{2}$. For any $s, t \in (0, T)$, we choose $\{t_i\}_{i=0}^N$ such that $t_0 = s, t_N = t$ and

$$t_i - t_{i-1} = \frac{t-s}{N} \quad (\text{A.13})$$

for all integers $1 \leq i \leq N$. Note that (A.11)–(A.13) imply that for any $0 \leq i \leq N-1$,

$$t_{i+1} - \theta R^2 \geq s - \theta R^2 = s - R^2 - \frac{t-s}{N} \geq \frac{s}{4} > 0$$

and

$$|p_{i+1} - p_i| = \frac{|x-y|}{N} \leq \frac{l}{N} = \frac{R}{2}.$$

Therefore, for any $0 \leq i \leq N-1$ we have $(t_{i+1} - \theta R^2, t_{i+1}) \subset (0, T)$ and $p_{i+1} \in B_{\frac{R}{2}}(p_i)$. Applying Theorem A.2 on $B_R(p_i) \times (t_{i+1} - \theta R^2, t_{i+1}) \subset \Omega \times (0, T)$, we have

$$u(p_i, t_i) \leq C u(p_{i+1}, t_{i+1}), \quad (\text{A.14})$$

where C depends only on c, n, μ and $\frac{1}{\theta-1} = \frac{R^2 N}{t-s}$. Here we used the fact that $t_i = (t_{i+1} - \theta R^2) + R^2$. Therefore,

$$u(y, s) = u(p_0, t_0) \leq C^N u(p_N, t_N) = C' u(x, t) \quad (\text{A.15})$$

where the constant C' in (A.15) depends only on $c, n, \mu, \min\{s, \delta^2\}, l, T_1$ and T_2 . The theorem is proved. $\square$

A direct corollary of Theorem A.3 is the following result.

Theorem A.4 *Let Ω be a bounded domain in $\mathbb{R}^n$. Suppose that $u(x, t) \in W_{n+1}^{1,2}(\Omega \times (0, T))$ is a nonnegative solution to the equation*

$$Lu = -\frac{\partial u}{\partial t} + a^{ij}(x, t)u_{ij} + b^i(x, t)u_i + cu = 0, \quad (\text{A.16})$$

where c is a constant and the coefficients satisfy (A.2)–(A.3) for a constant $\mu > 0$, and Ω', Ω'' are subdomains in Ω satisfying the following properties:

- (1). $\Omega' \subset \Omega'' \subset \Omega$, and Ω'' has a positive distance $\delta > 0$ from the boundary of Ω ;
- (2). Ω' can be covered by k balls with radius r , and all balls are contained in Ω'' .

Then for any s, t satisfying $0 < s < t < T$ and any $x, y \in \Omega'$, we have

$$u(y, s) \leq C u(x, t), \quad (\text{A.17})$$

where C depends only on $c, n, \mu, \min\{s, \delta^2\}, t - s, r$ and k .

Proof By the assumption, we can find finite many points $\mathcal{A} = \{q_1, q_2, \dots, q_k\}$ such that

$$\Omega' \subset \cup_{q \in \mathcal{A}} B_r(q) \subset \Omega''. \quad (\text{A.18})$$

For any $x, y \in \Omega'$, there exists two points in $\mathcal{A}$, which we denote by q_1 and q_2 , such that $x \in B_r(q_1)$ and $y \in B_r(q_2)$. Then x and y can be connected by a polygonal chain γ , which consists of two line segments $\overline{xq_1}, \overline{yq_2}$ and a polygonal chain with vertices in $\mathcal{A}$ connecting q_1 and q_2 . Clearly, the number of the vertices of γ is bounded by $k + 2$ and the total length of γ is bounded by $(k + 2)r$. Moreover, by the assumption we have $\gamma \subset \Omega''$ and each point in γ has a positive distance at least $\delta > 0$ from the boundary of Ω .

Assume that the polygonal chain γ has consecutive vertices $\{p_0, p_1, \dots, p_N\}$ with $p_0 = y, p_N = x$ and $1 \leq N \leq k + 2$. We apply Theorem A.3 for each line segment $\overline{p_i p_{i+1}}$ and the interval $[t_i, t_{i+1}]$, where $\{t_i\}$ is chosen as in (A.13). Note that

$$\frac{t - s}{k + 2} \leq t_{i+1} - t_i = \frac{t - s}{N} \leq t - s.$$

Thus, for any $0 \leq i \leq N - 1$ we have

$$u(p_i, t_i) \leq C u(p_{i+1}, t_{i+1}), \quad (\text{A.19})$$

where C depends only on $c, n, \mu, \min\{s, \delta^2\}, r, k$ and $t - s$, and (A.19) implies (A.17). This finishes the proof of Theorem A.4. $\square$

Appendix B The interior estimates of parabolic equations

In this ‘‘Appendix’’, we present the interior estimates of parabolic equations from G. Lieberman’s book [37] for the reader’s convenience.

Let $X = (x, t)$ be a point in $\mathbb{R}^{n+1}$ and $x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$. The norms on $\mathbb{R}^n$ and $\mathbb{R}^{n+1}$ are given by

$$|x| = \left(\sum_{i=1}^n (x^i)^2 \right)^{\frac{1}{2}}, \quad |X| = \max\{|x|, |t|^{\frac{1}{2}}\}.$$

Let Ω be a domain in $\mathbb{R}^{n+1}$. Let $d(X, Y) = \min\{d(X), d(Y)\}$ where $d(X) = \text{dist}(X, \mathcal{P}\Omega \cap \{t < t_0\})$. Here $\mathcal{P}\Omega$ denotes the parabolic boundary of Ω . We define

$$|f|_0 = \sup_{\Omega} |f|.$$

If $b \geq 0$, we define

$$|f|_0^{(b)} = \sup_{X \in \Omega} d(X)^b |f(X)|.$$

If $a = k + \alpha > 0$ and $a + b \geq 0$, where k is a nonnegative integer and $\alpha \in (0, 1]$, we define

$$\begin{aligned} [f]_a^{(b)} &= \sup \left\{ \sum_{|\beta|+2j=k} d(X, Y)^{a+b} \frac{|D_x^\beta D_t^j f(X) - D_x^\beta D_t^j f(Y)|}{|X - Y|^\alpha} : \right. \\ &\quad \left. X \neq Y \text{ in } \Omega \right\}, \\ \langle f \rangle_a^{(b)} &= \sup \left\{ \sum_{|\beta|+2j=k-1} d(X, Y)^{a+b} \frac{|D_x^\beta D_t^j f(X) - D_x^\beta D_t^j f(Y)|}{|X - Y|^{1+\alpha}} : \right. \\ &\quad \left. X \neq Y \text{ in } \Omega, x = y \right\}, \\ |f|_a^{(b)} &= \sum_{|\beta|+2j \leq k} |D_x^\beta D_t^j f|_0^{(|\beta|+2j+b)} + [f]_a^{(b)} + \langle f \rangle_a^{(b)}. \end{aligned}$$

We also define $|f|_a^* = |f|_a^{(0)}$, and define the spaces

$$H_a^* = \{f : |f|_a^* < \infty\}, \quad H_a^{(b)} = \{f : |f|_a^{(b)} < \infty\}.$$

With these notations, we have the following result.

Theorem B.1 (Theorem 4.9 of [37]) *Let Ω be a bounded domain in $\mathbb{R}^{n+1}$, and let $a^{ij} \in H_\alpha^{(0)}$ and $b^i \in H_\alpha^{(1)}$ satisfy*

$$\lambda |\xi|^2 \leq a^{ij} \xi_i \xi_j \leq \Lambda |\xi|^2, \quad [a^{ij}]_\alpha^{(0)} \leq A, \quad (\text{B.1})$$

$$|b^i|_{\alpha}^{(1)} \leq B \quad (\text{B.2})$$

for some constants A, B, λ and Λ . Let $c \in H_{\alpha}^{(2)}$ satisfy

$$|c|_{\alpha}^{(2)} \leq c_1 \quad (\text{B.3})$$

for some constant c_1 and let $f \in H_{\alpha}^{(2)}$. If $u \in H_{2+\alpha}^*$ is a solution of

$$-\frac{\partial u}{\partial t} + a^{ij}u_{ij} + b^i u_i + cu = f \quad (\text{B.4})$$

in Ω , then there is a constant C determined only by A, B, c_1, n, λ and Λ such that

$$|u|_{2+\alpha}^* \leq C(|u|_0 + |f|_{\alpha}^{(2)}). \quad (\text{B.5})$$

References

1. Bamler, R.H., Zhang, Q.S.: Heat kernel and curvature bounds in Ricci flows with bounded scalar curvature. *Adv. Math.* **319**, 396–450 (2017)
2. Brendle, S., Huisken, G.: Mean curvature flow with surgery of mean convex surfaces in $\mathbb{R}^3$. *Invent. Math.* **203**(2), 615–654 (2016)
3. Chen, B.L., Yin, L.: Uniqueness and pseudolocality theorems of the mean curvature flow. *Commun. Anal. Geom.* **15**(3), 435–490 (2007)
4. Cooper, A.: A characterization of the singular time of the mean curvature flow. *Proc. Am. Math. Soc.* **139**(8), 2933–2942 (2011)
5. Choi, H., Schoen, R.: The space of minimal embeddings of a surface into a three-dimensional manifold of positive Ricci curvature. *Invent. Math.* **81**(3), 387–394 (1985)
6. Chen, X.X., Sun, S., Wang, B.: Kähler Ricci flow, Kähler Einstein metric, and K -stability. [arXiv:1508.04397](https://arxiv.org/abs/1508.04397). To appear in *Geometry and Topology*
7. Chen, X.X., Wang, B.: Space of Ricci flows I. *Commun. Pure Appl. Math.* **65**(10), 1399–1457 (2012)
8. Chen, X.X., Wang, B.: Space of Ricci flow II. [arXiv:1405.6797](https://arxiv.org/abs/1405.6797)
9. Chen, X.X., Wang, B.: Space of Ricci flow II—Part A: moduli of singular Calabi-Yau spaces. *Forum Math. Sigma.* **5**, e32 (2017)
10. Chen, X.X., Wang, B.: Space of Ricci flow II—Part B: weak compactness of the flows. To appear in *Journal of Differential Geometry*
11. Chen, X.X., Wang, B.: Remarks of weak compactness along Kähler Ricci flow. [arXiv:1605.01374](https://arxiv.org/abs/1605.01374). To appear in *Proceedings of the seventh International Congress of Chinese Mathematicians*
12. Chen, X.X., Wang, B.: On the conditions to extend Ricci flow(III). *Int. Math. Res. Not. IMRN* **10**, 2349–2367 (2013)
13. Colding, T.H., Minicozzi II, W.P.: A course in minimal surfaces. *Graduate Studies in Mathematics*, 121. American Mathematical Society, Providence, RI (2011). xii+313 pp
14. Colding, T.H., Minicozzi II, W.P.: Embedded minimal surfaces without area bounds in 3-manifolds. *Geometry and topology: Aarhus* (1998), 107–120, *Contemporary Mathematics*, 258, American Mathematical Society, Providence, RI, 2000
15. Colding, T.H., Minicozzi II, W.P.: Smooth compactness of self-shrinkers. *Comment. Math. Helv.* **87**(2), 463–475 (2012)

16. Colding, T.H., Minicozzi II, W.P.: Generic mean curvature flow I: generic singularities. *Ann. of Math.* **175**(2), 755–833 (2012)
17. Colding, T.H., Minicozzi II, W.P.: The singular set of mean curvature flow with generic singularities. *Invent. Math.* **204**(2), 443–471 (2016)
18. Ecker, K.: *Regularity Theory for Mean Curvature Flow*. Progress in Nonlinear Differential Equations and Their Applications, vol. 57. Birkhäuser, Boston (2004)
19. Ecker, K., Huisken, G.: Mean curvature evolution of entire graphs. *Ann. of Math.* **130**(3), 453–471 (1989)
20. Ecker, K., Huisken, G.: Interior estimates for hypersurfaces moving by mean curvature. *Invent. Math.* **105**(3), 547–569 (1991)
21. Gulliver, R., Lawson, B.: The structure of stable minimal hypersurfaces near a singularity. *Proc. Symp. Pure Math.* **44**, 213–237 (1986)
22. Hamilton, R.S.: Three-manifolds with positive Ricci curvature. *J. Differ. Geom.* **17**(2), 255–306 (1982)
23. Haslhofer, R., Kleiner, B.: Mean curvature flow of mean convex hypersurfaces. *Commun. Pure Appl. Math.* **70**(3), 511–546 (2017)
24. Haslhofer, R., Kleiner, B.: Mean curvature flow with surgery. *Duke Math. J.* **166**(9), 1591–1626 (2017)
25. Huisken, G.: Flow by mean curvature of convex surfaces into spheres. *J. Differ. Geom.* **20**(1), 237–266 (1984)
26. Huisken, G.: Asymptotic behavior for singularities of the mean curvature flow. *J. Differ. Geom.* **31**(1), 285–299 (1990)
27. Huisken, G., Sinestrari, C.: Mean curvature flow singularities for mean convex surfaces. *Calc. Var. Partial Differ. Equ.* **8**(1), 1–14 (1999)
28. Huisken, G., Sinestrari, C.: Convexity estimates for mean curvature flow and singularities of mean convex surfaces. *Acta Math.* **183**(1), 45–70 (1999)
29. Huisken, G., Sinestrari, C.: Mean curvature flow with surgeries of two-convex hypersurfaces. *Invent. Math.* **175**(1), 137–221 (2009)
30. Ilmanen, T.: Singularities of mean curvature flow of surfaces, preprint. <http://www.math.ethz.ch/~papers/pub.html> (1995)
31. Krylov, N., Safonov, M.: A certain property of solutions of parabolic equations with measurable coefficients. *Izv. Akad. Nauk SSSR Ser. Mat.* **44**(1), 161–175 (1980)
32. Krylov, N., Safonov, M.: A certain property of solutions of parabolic equations with measurable coefficients. *Math. USSR-Izv.* **16**(1), 151–164 (1981)
33. Le, N.Q., Sesum, N.: The mean curvature at the first singular time of the mean curvature flow. *Ann. Inst. H. Poincaré Anal. Non Linéaire* **27**(6), 1441–1459 (2010)
34. Le, N.Q., Sesum, N.: On the extension of the mean curvature flow. *Math. Z.* **267**(3–4), 583–604 (2011)
35. Le, N.Q., Sesum, N.: Blow-up rate of the mean curvature during the mean curvature flow and a gap theorem for self-shrinkers. *Commun. Anal. Geom.* **19**(4), 633–659 (2011)
36. Lin, L.Z., Sesum, N.: Blow-up of the mean curvature at the first singular time of the mean curvature flow. *Calc. Var. Partial Differ. Equ.* **55**(3), 16 (2016). Art. 65
37. Lieberman, G.M.: *Second Order Parabolic Differential Equations*. World Scientific Publishing Co., Inc., River Edge (1996). xii+439 pp
38. Mantegazza, C.: *Lecture Notes on Mean Curvature Flow*, Progress in Mathematics, 290. Birkhäuser/Springer Basel AG, Basel (2011). xii+166 pp
39. Sesum, N.: Curvature tensor under the Ricci flow. *Am. J. Math.* **127**(6), 1315–1324 (2005)
40. Simon, M.: Extending four dimensional Ricci flows with bounded scalar curvature. [arXiv:1504.02910](https://arxiv.org/abs/1504.02910)
41. Wang, B.: On the conditions to extend Ricci flow. *Int. Math. Res. Not. IMRN* 2008, no. 8, Art. ID rnn012, p. 30

42. Wang, B.: On the conditions to extend Ricci flow(II). *Int. Math. Res. Not. IMRN* **14**, 3192–3223 (2012)
43. Wang, L.: Geometry of two-dimensional self-shrinkers. [arXiv:1505.00133](https://arxiv.org/abs/1505.00133)
44. Wang, M.T.: The mean curvature flow smoothes Lipschitz submanifolds. *Commun. Anal. Geom.* **12**(3), 581–599 (2004)
45. White, B.: The size of the singular set in mean curvature flow of mean-convex sets. *J. Am. Math. Soc.* **13**(3), 665–695 (2000)
46. White, B.: The nature of singularities in mean curvature flow of mean-convex sets. *J. Am. Math. Soc.* **16**(1), 123–138 (2003)
47. White, B.: A local regularity theorem for mean curvature flow. *Ann. of Math.* **161**(3), 1487–1519 (2005)
48. White, B.: Curvature estimates and compactness theorems in 3-manifolds for surfaces that are stationary for parametric elliptic functionals. *Invent. Math.* **88**(2), 243–256 (1987)
49. White, B.: Lectures on Minimal Surface Theory. [arXiv:1308.3325](https://arxiv.org/abs/1308.3325)
50. Xu, H.W., Ye, F., Zhao, E.T.: Extend mean curvature flow with finite integral curvature. *Asian J. Math.* **15**(4), 549–556 (2011)
51. Zhang, Z.: Scalar curvature behavior for finite-time singularity of Kähler–Ricci flow. *Mich. Math. J.* **59**(2), 419–433 (2010)

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